Natural Capital and the Measurement of Productivity*

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Abstract

This paper investigates how omitting natural capital as an input to production affects estimates of productivity growth. We show benchmark measurements of productivity growth are strongly correlated with environmental change at the national level, consistent with the omission of natural capital biasing measurement. We provide theoretical conditions under which the measurement and inclusion of previously-omitted environmental inputs reduces bias and error when estimating productivity growth. Calibrated simulations using data from the World Bank suggest the inclusion of non-renewable natural capital will improve measurement of TFP for 96 percent of countries in our sample.

Keywords: Total Factor Productivity; National Accounts; Natural Capital; Measurement

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Innovation and technological change drove an unprecedented increase in economic activity over the past two centuries. Industrialization simultaneously muddled societal awareness of fundamental connections between humans and nature. Society's interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital. Theodore Roosevelt noted this phenomenon at the outset of the 20$^{\text{th}}$ century:

> "With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature." – Roosevelt

Formal systems of economic measurement have followed this trend; extant structures provide little guidance for separating nature's contribution to production from the contributions of other factors. Starting with Simon Kuznets' report to Congress in 1934, the broader development of statistical, economic, and environmental policy, has focused on measuring output and productivity growth by tracking capital and labor over time (Kuznets). While governments historically emphasized tracking natural resources for posterity and avoiding overuse, features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today. For example, sailors, vessels, and sonar are essential inputs to industrial fishing that are measured by national accountants on a regular basis. Equally important is the fish stock and its ability to replenish itself over time (Hannesson). However, the fish stock as an input to production is often excluded from economic measurement. This continues to be the case despite managed fish stocks (those within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts (European Commission and Development).

This paper examines how the omission of the environment as a factor in production affects a leading metric of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP is defined as the growth in output unexplained by growth in inputs, and it has been a benchmark metric for economy-wide technological progress and innovation (Solow). Changes in TFP over time play a crucial role in shaping governments' budget projections and policy agendas over long time horizons (Congressional Budget Office). When natural capital affects production, the omission of environmental assets from national accounting frameworks will induce error when estimating productivity. Failing to account for how changes in the environment determine output may then cause governments to overestimate the ability for society to grow without the use of nature or and misattribute the effects of economic policy (Nordhaus and Tobin; Brandt, Schreyer, and Zipperer).

The first half of this paper begins with a review of how the omission of inputs to the production function affects residual-based estimates of TFP and covers why natural capital is often omitted in practice. We give predictions for the correlation of environmental change with TFP growth when natural capital is incorrectly omitted from the set of measured inputs used to construct TFP. We test whether a benchmark series for TFP growth is orthogonal to proxies for natural services in a balanced panel of data for 117 countries between 1996 and 2019 and find measured growth is strongly correlated with changes in the modest set of natural capital stocks we consider. These correlations with natural capital remain present when we use alternative productivity measurements taken from the economics literature and those taken from government statistical agencies, and are robust to a variety of alternative definitions for the set of omitted environmental inputs we consider.

The second half of the paper characterizes when including natural capital aids measurement of TFP in theory and then considers whether it does so in practice. We begin by giving formal conditions under which additional measurement unambiguously reduces bias; the conditions are quite restrictive and unlikely to hold in most settings. As the theoretical conditions are not formally testable, we turn to simulation-based evidence to gauge the extent to which including natural capital improves measurement of TFP. Results from our simulations suggest the gains in measurement accuracy are substantial. Across simulations calibrated using data from 100 countries, the median level of improvement in measurement (measured as the reduction in mean squared error for estimated average TFP growth) is 6.6 basis points, or a little over 14 percent of average TFP growth at the country level during our sample period. For the case of the United States, compounding an error reduction of this magnitude over a standard (10-year) budget horizon would imply a forecast error for cumulative revenues of a little over $300 billion in 2025. This figure is two to three times larger than the annual year-on-year forecast error in deficit projections by the CBO, or roughly a quarter of the aggregate cost of compliance with the Clean Air Act between 1970 and 1990 (Congressional Budget Office; U.S. Environmental Protection Agency).

Related Literature: Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value (towards their broader measure of economic welfare) as well as the productive value of land (Nordhaus and Tobin). Subsequent work incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to produced capital (e.g., Mankiw, Romer, and Weil; Caselli; Caselli and Feyrer; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis). These studies apportion capital income between returns on physical investment and resource rents in order to correct for the effects of the latter when comparing the marginal product of produced capital and its share of national income across countries (Caselli and Feyrer; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis). Broadly, the literature has highlighted the importance of correcting for resource rents when calculating the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains high (Hsieh and Klenow). We add to this literature in characterizing explicitly how introducing previously-omitted factors of production has ambiguous effects on the bias in measurement of output elasticities and factor shares.

A nascent strain of literature focuses explicitly on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to that of Brandt, Schreyer, and Zipperer, who were the first to study the effects of adjusting existing TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuel. The authors show these adjustments depend on the relative rate of growth in the use of non-renewable resources to that of changes in other inputs, which leads measurement of natural capital to have ambiguous effects on growth estimates. Bastien-Olvera and Moore perform a similar exercise when calibrating a regional integrated assessment model used for adjusting the social cost of carbon to account for non-market ecosystem services. Finally, Freeman, Inklaar, and Diewert highlight the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular constructions of TFP (e.g., versus those in the Penn World Tables). We add to this literature by explicitly characterizing the conditions when measurement of non-renewable resources reduces errors in estimated TFP, as opposed to calculating the magnitude of the adjustments induced by inclusion. Our approach is also generalizable to considering the effects of including other potentially omitted capital stocks, such as renewable natural capital.

The remainder of the paper proceeds in three steps. Section 1 tests whether existing TFP metrics are orthogonal to proxies for environmental inputs. We show changes in a modest subset of natural capital stocks are strongly correlated with benchmark series for productivity taken from the academic literature and national statistical agencies, consistent with the notion that relevant environmental inputs are omitted. Section 2 derives sharp conditions under which including a partial accounting of natural capital improves measurement of TFP. Section 3 uses data on natural resources from the World Bank's Changing Wealth of Nations project to simulate how including non-renewable resources when measuring TFP affects in-sample error. Section 4 concludes.

1 Residual-Based Productivity Measurement

Neoclassical economics models output as attributable to two sources: factors of production (factor inputs) and technology. Final output $Y(t)$ is determined by the level of factor inputs (the vector $\boldsymbol{X}(t)$) and a summary statistic for the level technology available to society, total factor productivity, $TFP(t)$:

$$Y(t) \triangleq TFP(t) \times F(\boldsymbol{X}(t)). \tag{1}$$

Total factor productivity (TFP) is a theoretical modeling device; an index number for how technology determines the production possibility frontier at a given point in time.[1] It is a parsimonious treatment for how forces such as innovation and creative destruction shift what society can produce

[1]We assume throughout this paper that technical change is Hicks neutral (i.e., follows 1). A more general formulation of technical change from a modeling standpoint is the partial derivative of the function determining aggregate output in time.

given available resources (Romer; Aghion and Howitt). Under standard assumptions for the production function (e.g., the Cobb-Douglas or transcendental logarithmic forms), growth in output is given by

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \boldsymbol{X}(t)} \gamma_x(t) g_x(t), \tag{2}$$

where $g_{\cdot}(t)$ terms denote growth rates and $\gamma_x(t)$ are the output elasticities for each input.[2] The growth accounting equation in (2) states that all changes in output are attributable to factors of production or growth in TFP (Barro). Rearranging the growth accounting equation gives

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \boldsymbol{X}(t)} \gamma_x g_x(t); \tag{3}$$

TFP growth is the output growth unexplained by growth in inputs (Comin). The accounting identity in (2) imposes these two concepts of productivity growth—the growth rate of an index number for technology and the residual portion of growth unexplained by factor inputs—must coincide.

In practice, TFP growth is constructed as the portion of output growth not explained by growth in a set of *observed* factor inputs, which we group together as the set $\boldsymbol{I}$. Factor inputs (e.g., produced capital and labor) and final output are measured by national statistical agencies at regular frequencies, allowing for statisticians to take versions of (3) to the data. Residual changes in observed output—those unexplained by growth in observed inputs—are called the Solow residuals (Solow) and are imputed to productivity growth by solving

$$g_A(t) \triangleq \hat{g}_Y(t) - \sum_{i \in \boldsymbol{I}} \hat{\gamma}_i(t) \hat{g}_i(t), \tag{4}$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are estimates of growth and output elasticities, respectively. Versions of equation (4) are used by statistical agencies like the U.S. Bureau of Labor Statistics (Meyer and Harper) and the European Statistical Office (European Commission and Eurostat) to construct measures of productivity growth in national accounts, and in academic applications comparing productivity across countries and over time (Feenstra, Inklaar, and Timmer). Statistical agencies often label their versions of the estimand in (4) *multifactor productivity* (as opposed to TFP) to acknowledge they measures productivity growth beyond the factors they include (Organisation for Economic Co-operation and Development).[3]

If national accountants were able to calculate the dynamics of all factor inputs and their

[2]In Section 1 we refer to the $\gamma_x(t)$ terms as output elasticities. As shown in Appendix Section 1, all analysis holds if the $\gamma_x(t)$ terms are reinterpreted as the shares of factors of production in GDP when the production function $F(\boldsymbol{X})$ exhibits constant returns to scale and factor markets are perfectly competitive (Caselli and Feyrer).

[3]The OECD manual on productivity measurement actually states this explicitly when defining multifactor productivity: "... [t]his manual uses the MFP [multifactor productivity] acronym to signal a certain modesty with respect to the capacity of capturing all factors' contribution to output growth" (Organisation for Economic Co-operation and Development).

associated output elasticities, a natural candidate for $\boldsymbol{I}$ would be to include all observed stocks to ensure $\boldsymbol{X} \subseteq \boldsymbol{I}$. A goal then may be to measure all natural capital stocks and include them all at once (Nordhaus and Kokkelenberg). In the interim, the set of measured inputs $\boldsymbol{I}$ available to practitioners remains limited. Residual-based approaches to productivity measurement deployed by national statistical agencies and researchers typically do not include environmental inputs (Meyer and Harper; Feenstra, Inklaar, and Timmer; European Commission and Eurostat; European Commission and Development).[4] These omissions are not active decisions to ignore nature; instead, they reflect limited state capacity and the difficulty in measuring natural capital versus other factors of production. Measuring changes in the flows of services from natural capital (c.f., the number of cycles from a turbine in the case of produced capital or total hours worked in the case of labor) that enter the production function each year is difficult; rates of extraction are often not summary statistics for the flow of services the stocks create (e.g., the fishery example in the introduction). Measurement is further complicated by the difficulty of attributing factor income generated by most forms of natural capital. Many services provided by the environment are public goods (e.g., climate and atmosphere) where the private cost of use is zero, rendering income-based approaches to measuring factor shares infeasible.

To illustrate the effects of omitting natural capital inputs when estimating TFP, consider a version of this economy with four factor inputs: produced capital, labor, non-renewable natural capital, and renewable natural capital, such that $\boldsymbol{X}(t) = \{K(t), L(t), N_1(t), N_2(t)\}$. Suppose a national accountant includes only two inputs, produced capital and labor, when estimating TFP for this economy. Conditional on her choice of inputs ($\boldsymbol{I}$) and her estimates of factor input growth rates and output elasticities ($\hat{\boldsymbol{g}}, \hat{\boldsymbol{\gamma}}$), the Solow residual can be decomposed as

$$g_A(t)|\boldsymbol{I}, \hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x \in \{N_1, N_2\}} \gamma_x(t) g_x(t)}_{\text{Omitted natural capital inputs}} + \underbrace{\sum_{i \in K, L} \left[\gamma_i(t) g_i(t) - \hat{\gamma}_i(t) \hat{g}_i(t)\right]}_{\text{Measurement error}}. \tag{5}$$

When natural capital inputs are omitted and $\boldsymbol{X} \setminus \boldsymbol{I} \neq \emptyset$, (3) is a biased estimator for TFP growth; the contributions of the omitted natural capital inputs are misattributed to TFP because they are spuriously absorbed by the Solow residual.[5] Correlation between the Solow residual in (5) with changes in a given natural capital stock (or other potentially omitted input) over time is evidence that the stock in question may be an omitted input in production. Conversely, a precise null of zero correlation between a natural capital stock and estimated TFP growth is suggestive that the empirical growth accounting equation contains the true set of factor inputs (i.e., $\boldsymbol{X} \subseteq \boldsymbol{I}$) and omitting the candidate stock for inclusion is unlikely to bias measures of TFP growth.

[4] At best, some TFP growth estimates adjust for payments to environmental factors that produce marketed goods and services where factor income is observed in government accounts. A unified framework would have to account for non-market services, such as the absorption of man-made pollution, when accounting for the totality of inputs into producing all goods and services humanity consumes.

[5] It is possible measurement error exactly cancels out the omitted terms and the estimator is unbiased even when inputs are omitted. We view this case as unlikely.

1.1 Testing for Omitted Natural Capital Inputs

The decomposition in (5) shows there are three potential channels which may cause the residual-based TFP estimator residual to be correlated with natural capital: (1) correlation between natural capital use and unobservable growth in TFP, (2) correlation with omitted natural capital stocks that affect output (i.e., are elements of $\boldsymbol{X}$), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While an econometrician cannot distinguish among these three potential sources of correlation, she can test whether the Solow residuals are uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with measured TFP growth, then we cannot rule out channel (2). Finding this form of empirical correlation in the data suggests that natural capital is more likely to play a role in output.

In a world with perfect information over the true growth rates of factor inputs and TFP, we could test channel (2)—whether a given natural capital stock is an omitted variable—in isolation. However, as we only observe g_A jointly as the sum of the three terms in (5), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. This leads to two potential sources of type one errors where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i} but spuriously attribute it to N_i being an omitted input in production. These cases correspond to channels (1) and (3); when g_{N_i} is systemically correlated with true innovation or measurement error in the growth accounting equation. This also brings about the potential for type two errors: when g_{N_i} is an omitted variable in the growth accounting equation and correlated with measurement error or TFP growth, it is possible these correlations are offsetting. The econometrician may then correctly fail to reject the null hypothesis of no correlation between g_A and g_{N_i}, but spuriously attribute this to natural capital playing no role in production.[6]

1.2 Estimating Equation

To test for correlations, we regress measured national TFP growth ($g_{A_{j,t}}$) from the Penn World Tables (PWT, Feenstra, Inklaar, and Timmer) on growth in the seven country-level natural capital stocks ($g_{N_{n,j,t}}$) measured by the 2024 version of the World Bank's The Changing Wealth of Nations (CWON) dataset. Equation (6) shows the empirical analog of (5) that we take to the available data. We estimate (6) on a balanced panel of 117 countries over the 1996-2019 period:[7]

[6] Several features of measurement may bring about these spurious results in practice. First, estimating output elasticities is nontrivial even for cases where factor income can be reasonably attributed (Caselli and Feyrer). Second, the lack of attributed factor income makes it difficult to delineate which changes in measured natural capital stocks constitute proxies for changes in flows entering production. Finally, while measures of some forms of innovation exist (e.g., patent filings and seed varietals, used as proxies for innovation in Acemoglu et al. and Moscona and Sastry respectively), underlying correlations between environmental change and TFP growth are not falsifiable.

[7] The CWON dataset in its raw form contains an unbalanced panel of natural capital stocks in each country covering 1995-2020. We impute missing values as zero changes from the last period where measurements were available leaving us with a balanced panel of growth in natural capital stocks between 1996 and 2020 (see Appendix Section 2.2 and 2.3 for a discussion of the imputation process for missing data along with robustness checks to ensure the results are not an artifact of this process). Data on total factor productivity levels in the PWT version 10.01

$$g_{A_{j,t}} = \sum_{i \in \boldsymbol{I}} \beta_i g_{N_{i,j,t}} + \sum_{z \in \boldsymbol{Z}} \theta_z g_{z,j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \tag{6}$$

where j and t index countries and years respectively. The set $\boldsymbol{I}$ covers the seven natural capital stocks measured at the country level present in the CWON, all measured in physical quantity units: urban land, agricultural land, timber land, non-timber forests, mangroves, fishery biomass, and hydroelectric power generation.[8] Coefficients β_i govern their relationship with TFP growth. The summation over terms $g_{z_{j,t}}$ allows for specifications that control for growth in other factor inputs such as produced capital, employment, labor's share of income, and human capital taken directly from the PWT (Feenstra, Inklaar, and Timmer). Terms (δ_j, ζ_t) allow for country and year fixed effects (labeled TWFE in Table 1), and $\varepsilon_{j,t}$ are error terms capturing other sources of variation in TFP growth.

Our estimands of interest (the β_i terms) are not the output elasticities associated with various natural capital stocks. Inference on these coefficients does not test for causal effects of changes in natural capital stocks on productivity growth, and they do not separately identify the omitted variable bias we seek to test from other potential channels for correlation. These coefficients identify the (conditional) covariance between growth in natural capital stocks and growth in measured TFP. Failing to reject the joint null hypothesis of interest:

$$H_0 : \boldsymbol{\beta} = \mathbf{0} \tag{7}$$

indicates only that we cannot rule out that the set of natural capital stocks we consider is uncorrelated with the combination of components in the decomposition of TFP growth in (5).

Three assumptions must hold in order to interpret the rejection of the null hypothesis in (7) as evidence that natural capital inputs are omitted when calculating TFP: (1), measurement error in the (weighted) factor input growth terms used to construct estimates of TFP is uncorrelated with growth in natural capital stocks; (2), changes in natural capital stocks are a good proxy for changes in the flow of services they produce; and (3), the rate of true technical change g_{TFP} is conditionally uncorrelated with the changes in environmental inputs to production that we consider. These assumptions are strong; however, their failure would not be indicative of natural capital being irrelevant for national accountants. If assumption 1 fails, better measurement of natural capital will improve the measurement of factor inputs that currently enter national accounts even if it has no direct value when calculating productivity growth. If assumption 2 fails, it is a powerful argument for adjusting the natural capital stocks we measure in order to keep better track of the aspects of the environment which are important for the economy. Finally, if assumption 3 fails, measurement of natural capital would be a first-order priority for national accountants in order to determine how nations may adjust the use of the environment to accelerate true productivity

only cover years out to 2019, as such our estimating sample covers 1996-2019.

[8] A full description of how the data on individual natural capital stocks are constructed in the CWON and how we clean these data prior to estimation along with the list of countries covered in our analysis can be found in Appendix 2.2.

growth. Explicitly proving the failure of any of the assumptions above would instead bolster the importance of measuring natural capital independent of how it would help measure productivity growth.

1.3 Reduced-Form Results

Table 1 shows the estimates of $\hat{\beta}$ from various specifications of (6), each of which gives how a change in the growth of a given natural capital stock affects expected growth in TFP. Each column gives two tests of the null hypothesis in (7) that the vector of coefficients on growth in natural capital $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_7]'$ is the zero vector. Failing to reject H_0 suggests that the collection of renewable natural capital stocks in CWON plays no role in the production function, and omitting them is unlikely to bias current metrics of TFP.

The bottom rows of table 1 show p-values from two Wald tests of this joint null hypothesis across all seven natural capital stocks we consider. The first row constructs test statistics using asymptotic standard errors (clustered at the country level), while the second uses a semi-parametric wild bootstrap approach (Wu). Across all specifications, we can reject that the natural capital stocks we consider are uncorrelated with productivity growth at traditional significance levels under the tests using asymptotic standard errors, and the tests using the wild bootstrap approach are always at least statistically significant at the 10% level. In addition, almost all point estimates are positive, the expected sign if the natural capital stocks we consider are beneficial inputs to production and covary positively with output.[9] While the estimated joint effects of changes in natural capital appear precise, the overall residual variation explained by adding natural capital on the margin is limited. Including natural capital increases model R^2 by 0.006 (3 percent over baseline levels) for the benchmark fixed effects models in columns (3) and (4).

Robustness: Partial correlations and individual tests of coefficient precision suggest services from land are the environmental inputs most likely to be material for current TFP metrics, consistent with the suggestions for improving national accounting given in the latest edition of the PWT (Feenstra, Inklaar, and Timmer). To ensure missing land rents are not the sole drivers of the results here, we re-estimate equation (6) after excluding urban, agricultural, and timber land from the set of natural capital stocks. Under this specification focusing on a subsample of four renewable natural capital stocks outside of land, our estimates lose precision and we are only able to reject the null hypothesis at the 10 percent level in about half of the specifications we consider (see Appendix 2.3). This confirms the suggestions in Feenstra, Inklaar, and Timmer that better accounting for land rents in national accounts is likely a fruitful direction for expansion. Appendix 2.3.2 provides several additional checks of the robustness of our results to alternative choices for the set of natural capital stocks considered (e.g., excluding natural capital stocks with a high share

[9]The negative coefficient on forest cover may be due to increases coinciding with declines in logging or forestry activities which are counted in final output. This does not per se imply that forests are a drag on the economy, but is rather indicative that the services they produce (outside of timber) are not captured by measured output. More generally, any societal benefit produced by a given natural capital stock with no in market production will not cause a correlation with TFP growth in this exercise.

of missing or imputed values, expanding them to include non-renewable natural capital, excluding only urban land, and considering alternative sample periods). Appendix 2.3.3 examines the results for alternative choices of series for measured TFP (e.g., measures from EUROSTAT focusing on the European Union and agriculture-specific measures from the United States Food and Drug Administration).

2 The Marginal Inclusion of Potentially Omitted Inputs

Section 1 gives suggestive evidence that natural capital is an omitted input in existing measures of TFP. We now turn to examining when including changes in previously-omitted inputs in the growth accounting equation improves residual-based estimates of productivity growth. Intuition suggests including an omitted determinant of output in the growth accounting equation should improve estimates of TFP. However, as shown in in Appendix 1.4, it is only guaranteed under very strong assumptions. For intuition, consider when the residual-based estimator is applied to our four-input example economy. Suppose a national accountant is able to measure growth in output, labor, produced capital, and non-renewables as well as the associated output elasticities (or factor shares) without error, but cannot measure growth in renewable resources over time (g_{N_2}). If the accountant elects to omit non-renewables (g_{N_1}) when calculating TFP growth, substituting this into (4) yields the estimand

$$\hat{g}_A(t) \triangleq g_{TFP}(t) + \gamma_1 g_{N_1}(t) + \gamma_2 g_{N_2}(t) \tag{8}$$

for TFP growth. If the accountant instead includes non-renewables in the residual-based estimator, the estimand for TFP becomes

$$\tilde{g}_A(t) \triangleq g_{TFP}(t) + \gamma_2 g_{N_2}(t) \tag{9}$$

It is ambiguous whether (9) is a less biased estimate of g_{TFP} than (8) without making assumptions on the sign of growth in renewable inputs over time. Appendix 1.4 generalizes this result to the case where multiple inputs may be simultaneously omitted; bias in the residual-based estimator of average TFP growth is reduced only when all omitted terms share a sign in expectation. Criteria for improving in-sample metrics of goodness of fit (such as root mean squared error) are even more stringent, as they depend on higher moments for growth in the omitted stocks and the set of included inputs. The effects of including natural capital on the margin, even with perfect measurement of growth and estimation of output elasticities, are only guaranteed to improve estimates of TFP when it leads the estimator to be correctly specified (i.e., $\boldsymbol{X} \subseteq \boldsymbol{I}$) or the pairwise growth rates of omitted natural capital inputs share very similar covariance structures.[10]

The result that the marginal inclusion of natural capital may not improve estimates of TFP growth has not been shown previously in the literature. For example, existing work by Brandt,

[10]See Proposition 2 in Appendix 1.4

Schreyer, and Zipperer shows that under the assumption that non-renewable capital inputs are uncorrelated with the use of other natural capital stocks, TFP growth inclusive of non-renewable natural capital tends to be greater than when it is measured excluding non-renewable natural capital. However, because of the results shown here, it is not possible to know whether these reconstructed series that adjust for non-renewable resources bring estimates closer to true productivity growth over time.

3 Simulating the Effects of Incorporating Natural Capital when Estimating TFP

Section 2 examined the theoretical effects of reducing the number of omitted factor inputs on the performance of the Solow residual as an estimator of TFP. This section evaluates this question using the available data on growth in factor inputs, natural capital stocks, and output. We take a version of our running four-input economy to the data and examine whether including non-renewables on the margin improves residual-based estimates of TFP.

3.1 Setup

The goal of our exercise is to calibrate a version of the four-factor economy from Section 2 at the country level for our panel of countries in Section 1. We consider a case where a national accountant seeks to measure TFP growth for this economy, but measures only growth in the usage of produced capital, labor and non-renewable capital, N_1. We assume she suspects that services from renewable capital (N_2) are important inputs in production, but that she is unable to measure how their use has changed over time. Knowing this, she must choose whether to include growth in the use of non-renewable resources, g_{N_1}, when taking Equation (4) to the data to estimate TFP growth. Her criteria for including non-renewable resources in this thought experiment is whether it lowers expected in-sample mean squared error when estimating average TFP growth in her data, which we assume covers 25 years of growth in output and factor inputs.[11]

Data Generating Process for Output: Output growth in each country is comprised of growth in TFP, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vectors $\boldsymbol{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j. Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \boldsymbol{g}'_{j,t}\boldsymbol{\gamma}_j \tag{10}$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for that country.

[11]We round up to 25 from the size of 24 years in our estimating sample.

Calibrating Country-Level Growth: We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. We calibrate the mean ($\boldsymbol{\mu}_j$) and variance (Σ_j) governing the joint distribution of growth terms $\boldsymbol{g}_{j,t}$ in each country based on sample moments listed in Table 2. For natural capital, we follow the World Bank CWON methodology (World Bank) and construct aggregate Törnqvist quantity indices for non-renewable and renewable resource use at the country level over time (see Appendix Section 3). We set the parameters governing average growth in the use of natural capital equal to the sample means of growth in these indices between 1996 and 2019 (Table 2 rows one and two). Expected growth in produced capital services, labor, and TFP are set equal to their sample analogs in the Penn World Tables (Feenstra, Inklaar, and Timmer) over the same 1996-2019 period. Higher order moments in Σ are set based on the empirical covariances.

Structural Parameters: We set the output elasticity of services from renewable natural capital, $\gamma_2 = 0.05$ based on the analysis in Appendix Section 3.6. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al.), we set the output elasticity of services from non-renewable natural capital, γ_1, to 0.05. This choice is slightly larger than our point estimates for this output elasticity in Appendix Section 3.6, but in line with estimates of the factor share for fossil energy in the United States since 1947 (Hassler, Krusell, and Olovsson).

Procedure for Estimating TFP Growth: The national accountant observes 25 years' worth of data for a subset of growth in factor inputs—those for K, L, and N_1—along with growth in output in a given country. We assume she does not observe the distribution of national income earned across factors and thus cannot estimate output elasticities using factor shares. This ensures our thought experiment is consistent with estimating the Solow residual in empirical settings, where factor income generated by natural capital is not attributed to the environment in national accounts and unlikely to be observable for an econometrician.

To construct a measure of TFP growth that does not rely on growth in non-renewable resources, the national accountant estimates:

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \tag{11}$$

on time series of data for a given country using ordinary least squares (OLS). Her estimator for average TFP growth based only on growth in produced capital and labor is the intercept term $\hat{\gamma}_A$. We refer to this estimate as model 1. She estimates the analog for the case including non-renewable resources:

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \tag{12}$$

on time series for growth in $\{K, L, N_1\}$. The OLS-based estimator for average TFP growth inclusive of non-renewable capital is $\tilde{\gamma}_A$. We refer to this as model 2.

We then calculate the expected RMSE for $\hat{\gamma}_A$ and $\tilde{\gamma}_A$ in each country by evaluating them at the sample analogs of $\boldsymbol{\mu}_j$ and Σ_j relative to the calibrated values, $\mathbb{E}[g_{TFP}]$. This gives us a cross section of 100 countries for the effects of including natural capital when calculating TFP growth, which can be conditioned on observed levels and correlations between the use of natural capital over time at the national level.[12]

3.2 Results

Figure 1 plots the results from our calibrated thought experiment. Each cell overlain with an ISO3 code corresponds to the pair of empirical correlations between the growth of non-renewable capital inputs and produced capital inputs (x axis) and the growth of non-renewable and renewable capital inputs (y axis) we observe in the data for that country between 1996 and 2019. For each cell with an ISO code, the outcome is the change in the expected RMSE for estimated TFP growth when non-renewable capital is included on the margin:[13]

$$\Delta\text{RMSE}|\boldsymbol{g} \triangleq \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{g}]} - \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{g}]}d \tag{13}$$

A positive change in our RMSE metric indicates that including non-renewable resources improves the fit of a national accountant's estimate of TFP; a negative values indicates it worsens that fit. The z axis in Figure 1a plots these conditional changes in expected RMSE, $\Delta\text{RMSE}|\boldsymbol{g}$, for each country in our calibrated cross section; darker blue indicate where including non-renewables worsens measurement on the margin, while progressively lighter greens show the degree of improvement. The remaining values within the convex hull formed by the cross-section (dashed gray lines) are correlation pairs that do not correspond to the historical behavior of a specific country and are fitted using linear interpolation. The improvement surface then posits the (conditional) improvement for a hypothetical country when a national accountant includes non-renewable capital on the margin (while knowingly omitting renewable natural capital) when estimating TFP.

The most striking result from our simulation is given by the histogram in Figure 1b; for 96 out of the 100 countries for which we have sufficient data, including non-renewable resources improves the fit of estimated TFP growth. The median and mean improvements in measurement are sizable: 7 and 32 basis points respectively. As shown by the histogram in Figure 1b, these improvements are disperse (s.d. 65 basis points) and right skewed.[14] The magnitude of the median improvement is a little over 14% of average TFP growth across all countries during our sample period between 1996-2019. For the four countries where including non-renewables raises RMSE, the deterioration in fit is small (less than 1 basis point both on average and at the median).[15]

The gradient of the improvement surface in Figure 1a is suggestive of how the correlation between a candidate (potentially-omitted) input and other factors of production determines how much

[12]See Section 4.1 of the Appendix for the full procedure.

[13]The full derivation of this process is given in Appendix Section 4.

[14]The countries with largest improvements (in terms of how much including non-renewables reduces RMSE when estimating TFP growth) are Kenya, Laos, Israel, Tanzania, and Niger.

[15]The largest RMSE increase occurs for Chile, where RMSE rises by 3.3 basis points. Chile is an empirical case where the two components of bias in (11) (stemming from omitting two variables) happen to offset each other almost perfectly, causing the bias when only renewables are omitted in (12) to be larger in magnitude.

its inclusion affects the performance of estimated TFP. If the world looks increasingly like Kyrgyzstan or Cameroon—nations in the second quadrant where growth in the use of non-renewables is negatively correlated with produced capital but positively correlated with renewables—including non-renewable resources in natural accounts is more likely to improve estimates of TFP growth. In contrast, if the world becomes more like Chile—where growth in the use of non-renewables is positively correlated with produced capital, but negatively correlated with renewables—adding non-renewables to national accounts will have a limited impact on existing estimates of TFP growth.[16]

Figure 2 shows scatter plots illustrating the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in resource use and baseline levels of TFP growth. Panels 2a and 2b plot improvements in estimates of TFP growth against average growth in the use of non-renewable and renewable resources (g_{N_1} and g_{N_2}) at the country level. Of the 100 countries in our sample, 68 countries exhibit positive average growth in their use of non-renewables and 47 exhibit positive average growth in the use of renewable inputs. Growth in non-renewables tends to be about an order of magnitude larger than renewables.[17] Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process (set as the observed value in the PWT). Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. The median (mean) improvement is 10 (22) percent of baseline error when both natural capital stocks are omitted, and rises to more than 40% for over a quarter of the countries in the cross section of hypothetical outcomes. Most of this improvement is driven by a reduction in the bias of the estimators as opposed to a smaller expected variance; bias drives 98.5 percent of RMSE on average in model 1 (where non-renewables are excluded), and 81 percent of bias on average in model 2 (where non-renewables are included).[18] Additional cross sectional analysis examining how the country-level outcomes in our simulations relate to other factors that play a role in measuring productivity (e.g., statistical capacity and the share of GDP stemming from resource rents) can be found in Appendix 4.4.

The resulting improvement for most countries is of a magnitude that is policy-relevant. The median reduction in RMSE across the 100 countries in our sample is 6.6 basis points (1/100th of a percentage point). This translates to an annual forecast error of roughly 0.07 percentage point when historical estimates are used to forecast average growth in the future. Applying this median value to the United States (which has an RMSE reduction of 26 basis points in our simulation), when this forecast error is compounded over ten years (the standard CBO budget horizon) at current

[16]The correlations between non-renewable natural capital and produced capital (renewable natural capital) are jointly significant at standard levels when ΔRMSE is projected onto the elements of $\boldsymbol{\mu}_j$ and Σ_j that enter the RMSE calculations. See Appendices 4.2 and 4.3 for details and reduced-form analysis of other how other sample moments determine ΔRMSE.

[17]Figure 4 in Appendix 4.4 shows a version of these two panels for all 12 sample moments affecting ΔRMSE at the country level.

[18]See Appendix 4 for the derivation and decomposition of the changes in RMSE in terms of changes in bias and variance.

levels of output, forecast cumulative revenues would deviate from projections by approximately \$300 billion.[19] This forecast error is roughly one quarter of the aggregate social cost of compliance with the Clean Air Act in the United States between 1970 and 1990 in magnitude, one of the broadest environmental policies enacted globally to date (U.S. Environmental Protection Agency).

Comparison to existing results in the literature: The magnitude of changes in estimates of annual TFP growth are similar to those produced by Brandt, Schreyer, and Zipperer, who adjust existing productivity metrics for a subset of non-renewable resources considered here. The authors find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for a panel of 21 countries between 1986 and 2008. Freeman, Inklaar, and Diewert use an earlier version of the Changing Wealth of Nations dataset to examine how inclusion of subsurface minerals and fossil fuels affects estimates of TFP relative to that of the United States. They find omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies such as those in OPEC or those with large mining sectors, in-line with the proportional reductions in error we show in Figure 2f. Finally, our estimates for the median reduction in RMSE when estimating TFP growth is about one-third the magnitude of the latest revisions of TFP estimates (0.2 percentage point) made by the United States' Bureau of Labor Statistics for 2024 (U.S. Bureau of Labor Statistics).

Sensitivity Analysis: We present sensitivity analysis of our simulation results to alternative choices of output elasticities for natural capital and sample periods for moments governing input growth in Appendix 4.5. Appendix 4.5.1 performs a Monte Carlo analysis of our exercise across ten million alternative pairs of output elasticities drawn from an iid uniform distribution over the interval $\mathcal{U}[0, 0.1]$. The median (mean) RMSE reductions fall to 2 (28) basis points and are negative under about 30 percent of alternative parameterizations, highlighting the importance of better measurement and apportionment of income generated by natural capital to generate better estimates of their contributions to output. Appendix 4.5.2 shows that re-estimating the sample moments in Table 2 based on two partitioned subsamples of the original 1996-2019 data leads to larger estimates of RMSE improvements across the board and minimal qualitative changes. The set of countries for which these improvements are negative remains quite small in both cases (ten and four countries respectively in the early and late periods).

4 Conclusion

Fifty-four years after Nordhaus and Tobin, research continues to highlight new channels by which environmental change affects society and welfare. An increasing share of this research has focused on responding to high-level calls for improved welfare measures, e.g., the U.N. Secretary General's push for measures "Beyond GDP" (United Nations Statistics Division). The importance of such measures has not gone unrecognized by academic economists; there is a longstanding literature on how GDP could be adjusted to better account for the determinants of welfare beyond market

[19] See Appendix 5 for breakdown of the back-of-the-envelope calculation here.

activity (Fleurbaey; Jorgenson).[20] This paper gives evidence that measurement of natural capital benefit governments irrespective of goals beyond GDP. When natural capital affects production, omitting it from national accounts biases estimates of economic growth. This supports the original hypothesis by Nordhaus and Tobin in 1973 that leaving things out ("treating as free things that are not really free") of national accounts "gives the wrong signals for the directions of economic growth" in addition to affecting measurement of national welfare (Nordhaus and Tobin). While measuring the environment will remain important for assessing the sustainability of development, regulatory compliance, and public health, we show here systematically connecting changes in the environment to economic statistics will benefit existing measurement that is already used to guide policy.

More generally, the posterity value of measurement may be high in the future even for resources that lack a role in current production processes. Fenichel argues a critical feature of natural capital is understanding a small number of accounting boundaries, in part to separate resources currently entering production from those that are not. This insight applies beyond the environment; better measurement allows us to more-accurately apportion current income to today's productive resources and ensure it can be done in retrospect if our understanding of which factors drive aggregate production changes. Innovation itself, measured by TFP, will change what factors drive future innovation. Better measurement will help address this "new inputs" challenge in measuring the economy. Expanding the scope of economic measurement is substantially easier when done contemporaneously (c.f. when done retrospectively). As future generations develop new productive technologies that draw on previously-unused inputs, improved measurement will enable future generations to achieve commensurate progress in informing future fiscal forecasts.

[20] Conte, Addicott, and Millerhaller provide micro-founded support for this position with regard to the environment, demonstrating that reductions in natural capital stocks generate real income effects for consumers.

Table 1: TFP Growth vs. Growth in Renewable Natural Capital Stocks

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.301	0.303	0.205	0.215	0.149	0.060
	(0.095)	(0.098)	(0.067)	(0.075)	(0.062)	(0.118)
Timber Land	0.164	0.166	0.071	0.075	-0.003	0.026
	(0.090)	(0.075)	(0.074)	(0.071)	(0.068)	(0.104)
Agricultural Land	0.027	0.043	0.030	0.036	0.041	0.043
	(0.025)	(0.025)	(0.022)	(0.023)	(0.021)	(0.017)
Forest Cover	-0.067	-0.111	-0.005	0.012	0.083	-0.027
	(0.094)	(0.091)	(0.079)	(0.083)	(0.053)	(0.076)
Mangroves	0.122	0.067	0.357	0.327	0.318	-0.108
	(0.244)	(0.242)	(0.299)	(0.322)	(0.367)	(0.427)
Fishery Biomass	0.112	0.026	0.094	0.072	0.047	-0.104
	(0.098)	(0.077)	(0.108)	(0.101)	(0.135)	(0.108)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.007	0.003	0.007	0.005	0.002	0.039
Bootstrap Wald	0.089	0.056	0.044	0.018	–	0.037
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables enter as log changes. Standard errors clustered at the country level are shown in parentheses. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator and reports robust (as opposed to clustered) standard errors. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Standard errors for th Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis (Wu). No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 2: Simulation Moments and Sources

Population Moment	**Sample Moment**	**Source**
$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON
$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON
$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Σ	$\widehat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the covariance matrix Σ. CWON refers to the World Bank *Changing Wealth of Nations* database (World Bank). PWT refers to the Penn World Tables version 10.01 (Feenstra, Inklaar, and Timmer).

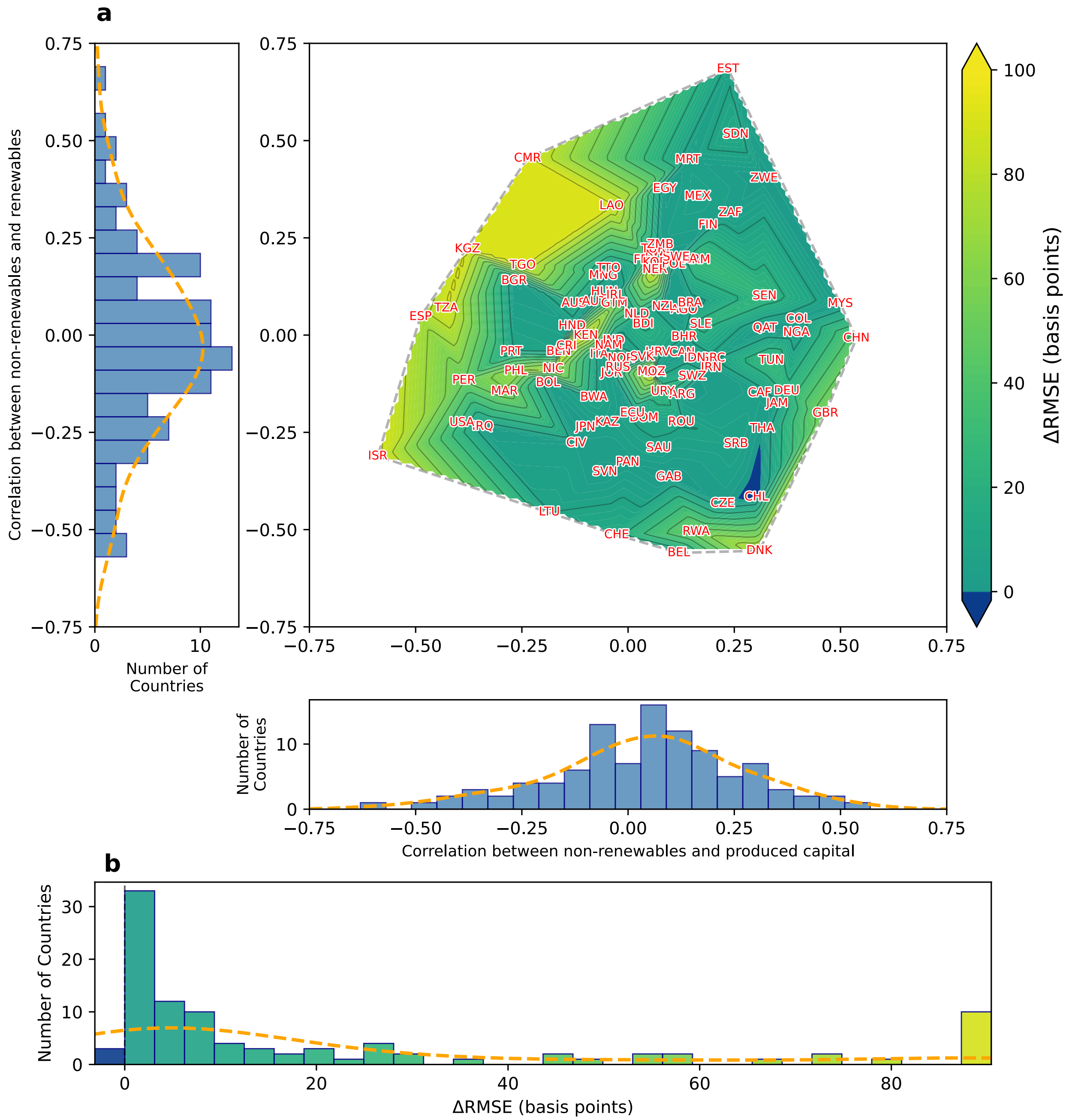

Figure 1: Panel 1a shows a contour plot of $\Delta\text{RMSE}|\boldsymbol{g}$ (measured in basis points) when adding g_{N_1} to the residual-based TFP estimator plotted against observed correlations between growth in non-renewables and produced (renewable) capital at the country level between 1996 and 2019 for the 100 countries we consider. Contour lines denote level sets of the reduction in RMSE. ISO are overlain on the simulated values for countries in our dataset while other points within the convex hull (dashed grey line) are fitted using piecewise linear interpolation after topcoding the z values at the 90^{th} percentile (90 basis points). Histograms on each axis show the empirical distributions of the correlations at the country level. Panel 1b shows a histogram of the reductions in mean squared error shown in Panel 1a with corresponding color coding for each bin. See Appendix Table 15 for a full list of country-level sample moments and changes in RMSE.

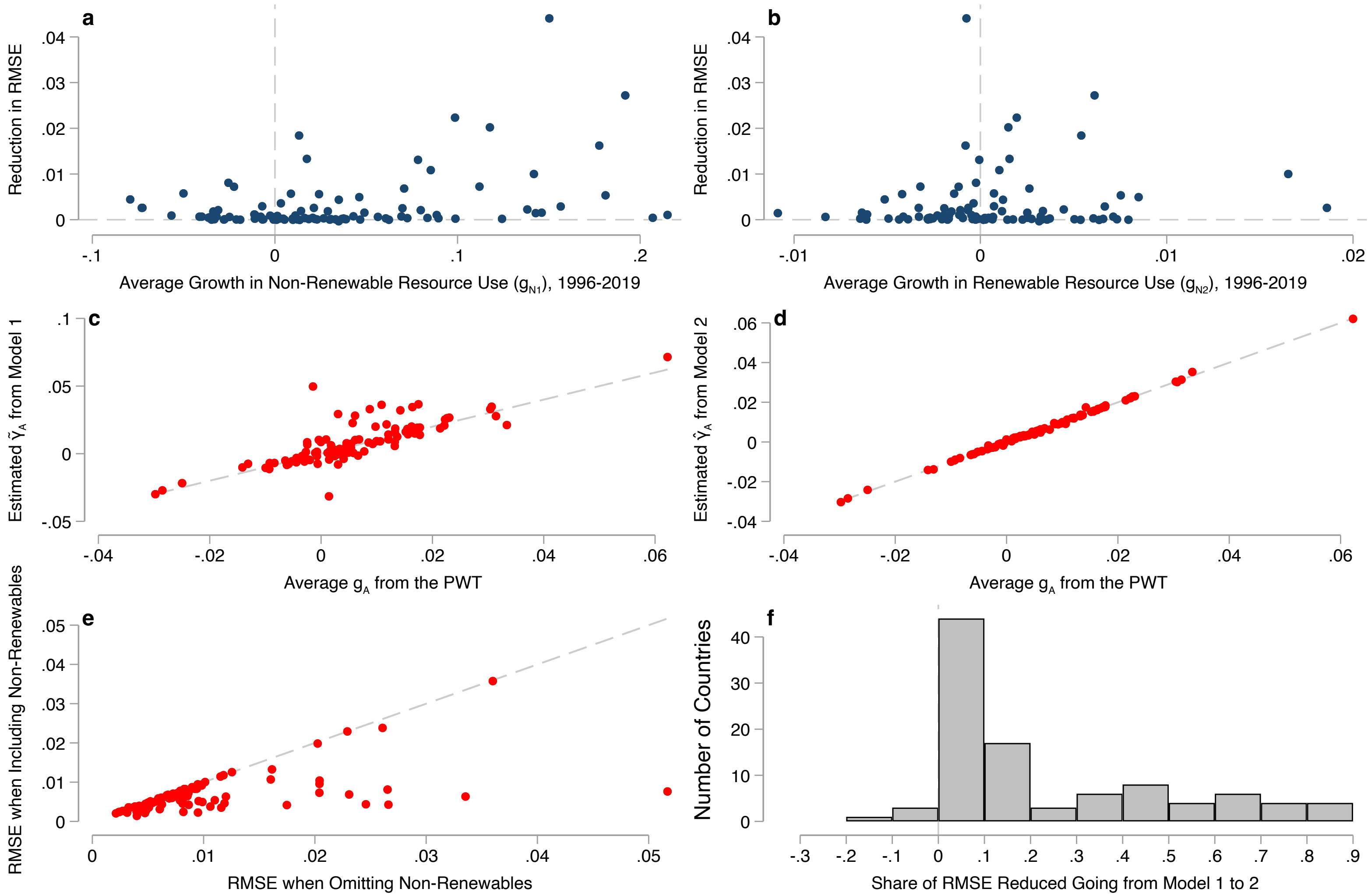

Figure 2: Panels 2a and 2b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 100 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 2a, 2b, and 2f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 2c-2e show the 45 degree line.

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Supplementary Information for Natural Capital and the Measurement of Productivity*

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Contents

1 The Theory of Natural Capital and Total Factor Productivity in Growth Accounting

This appendix examines the conditions under which incorporating a previously unmeasured natural capital input into a macroeconomic production function improves residual-based estimates of total factor productivity (TFP). Our laboratory for measurement is a theoretical economy in which output is governed by TFP as well as by the inputs to the aggregate production function F. We assume that F is log-linear in inputs and takes at least four factors as arguments: labor L, produced capital K, and $J > 1$ natural capital inputs indexed $\{N_{j=1}, N_{j=2}, \ldots, N_{j=J}\}$.[1] The logarithm of output at each instant t is given by

$$\begin{aligned} \ln(Y(t)) &= \ln\Big(TFP(t) \times F(K(t), L(t), \{N_j(t)\}_{j=1}^{J})\Big) \\ &= \ln(TFP(t)) + \gamma_K(t)\ln(K(t)) + \gamma_L(t)\ln(L(t)) + \sum_{j=1}^{J}\gamma_j(t)\ln(N_j(t)) \end{aligned} \tag{1}$$

where $\gamma_K, \gamma_L, \{\gamma_j\}_{j=1}^{J} > 0$ denote output elasticities with respect to the corresponding factor inputs. If F is homogeneous of degree one in its inputs (i.e., production exhibits constant returns to scale, CRS), output can be expressed as

$$Y(t) = TFP(t) \times \Big(F_K K(t) + F_L L(t) + \sum_{j=1}^{J} F_{N_j} N_j(t)\Big) \tag{2}$$

where F_x denotes the partial derivative of F with respect to argument x. Under the assumption that the output elasticities γ are time-invariant, the growth rate of output in (1) at a given time t can be expressed as[2]

$$\frac{\dot{Y}(t)}{Y(t)} = \frac{T\dot{F}P(t)}{TFP(t)} + \frac{\dot{F}(t)}{F(t)} = \frac{TF\dot{P}(t)}{TFP(t)} + \gamma_K\frac{\dot{K}(t)}{K(t)} + \gamma_L\frac{\dot{L}(t)}{L(t)} + \sum_{j=1}^{J}\gamma_j\frac{\dot{N}_j(t)}{N_j(t)} \tag{3}$$

where dots denote derivatives of a given variable with respect to time. Let $g_X(t) \triangleq \dot{X}(t)/X(t)$ denote the growth rate of variable X at time t for $X \in \{A, K, L, \{N_j\}_{j=1}^{J}\}$. Rearranging (3) yields the accounting-based formula for calculating TFP growth: g_{TFP} is output growth less an output-elasticity weighted sum of growth in inputs:

$$g_{TFP} = g_Y - \gamma_K g_K - \gamma_L g_L - \sum_{j=1}^{J}\gamma_j g_{N_j} \tag{4}$$

We refer to (4) as the *growth accounting* equation going forward. The growth accounting equation states that if output elasticities in the production function are known to the econometrician and

[1] Without loss of generality, other omitted inputs that are not natural capital can be subsumed into the set of J inputs beyond produced capital and labor.

[2] Equation (3) still holds when output elasticities depend on the level of inputs and/or time, but they must be time-indexed.

she observes growth in both inputs used and output, she can recover the full path of TFP growth.

Several barriers complicate the inclusion of natural capital (NK) stocks in empirical growth accounting exercises. First, a national accountant may not know the full set of natural capital stocks $\{N_j\}_{j=1}^{J}$ that enter the production function. National statistical agencies have not traditionally collected data on natural capital. The System of National Accounts (SNA) standard international guidance includes a set of natural assets, but few countries tabulate changes in these assets over time (European Commission and Development). The System of the Environmental Economic Accounts Central Framework and Ecosystem Accounts provides an alternative set of natural assets they recommend countries track (United Nations et al.˙. While some countries have begun collecting their own natural capital accounts, these data are unlikely to capture the full scope of natural assets that may enter the production function. Second, the accountant cannot (at least directly) observe output elasticities $\gamma_{\cdot}$. Elasticities must be estimated, which introduces uncertainty when these elasticities are used in the growth accounting equation. Third, the accountant may face measurement error when calculating growth in output and factor inputs as well as any other auxiliary measurement required to estimate output elasticities.

This section lays out a theoretical framework for examining how uncertainty over which natural capital stocks enter the production function affects estimates of output elasticities and TFP growth. We show conditions under which knowing more information about some, but not all, inputs sharpens estimates of TFP growth. These are policy-relevant questions; it is unlikely that national accountants will go from current practices to identifying the complete set of NK assets without first identifying an incomplete set. Further, budget and institutional constraints are unlikely to allow for the introduction of all NK assets at once. The theory here outlines conditions under which an intermediate step measuring some of the NK inputs yields contemporary improvements in TFP measurement.

1.1 Estimating Output Elasticities

Estimating TFP growth using the growth accounting method requires estimates of the output elasticities in (4). There are two general approaches that require distinct sets of assumptions. The first approach, the income approach, relaxes the assumption of log linearity and operates under two alternative assumptions: F is instead assumed to exhibit constant returns to scale (as in 2), and that the market structures for factor inputs as well as intermediate and final goods are known. The second approach, the regression approach, requires only that the aggregate production function in the economy is log-linear as in (1). This section touches briefly on the income approach before giving an overview of the regression approach as well as its properties when natural capital is omitted from the growth accounting equation.

1.2 Income-Based Approach

When the production function F exhibits CRS and factor markets are perfectly competitive, the shares of national income accruing to each factor input are equal to their output elasticities in equilibrium (Caselli and Feyrer). For the case of capital, this implies

$$\gamma_K = \frac{F_K K}{Y} = \frac{RK}{Y} \tag{5}$$

The ratio of total rents paid to capital, RK, divided by output Y, which is usually measured as gross domestic product (GDP), is an unbiased estimator for the capital output elasticity in the economy. This procedure is especially useful given that its arguments can be measured in *nominal* terms (i.e., using money units) as opposed to real terms. Analogous procedures can be used to estimate elasticities for labor and natural capital factors where the factor income accruing to them is observable. An added advantage is that under the CRS assumption, factor shares of national income must sum to one.

These estimators become biased if any factor income attributable to an omitted factor (e.g., any natural capital stock that enters production but is not paid its factor income) is misattributed to observed factors. Manning, Taylor, and Wilen illustrate how this case is likely for industries that use rival public goods as inputs (the authors there use a fishery as a running example) and show how it complicates measurement of factor shares in production. The estimator in (5) is also biased if the aggregate production function does not exhibit CRS or the econometrician does not know the structure governing the market for factor inputs. The challenges of estimating factor shares using this method are well known (Caselli and Feyrer; Hsieh and Klenow; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis). The resulting measurement error when using the income-based approach to calculate TFP growth depends on how national income is measured, how national is attributed across the factors of production, and how the error propagates through to TFP calculations.

1.3 Regression-Based Estimators

The regression based approach relies on the log-linear restriction in (1). When production is Cobb Douglas or well-approximated by its Taylor expansion such as in the translog case, the first two moments of the stochastic process governing input and output growth is a sufficient statistic for the set of output elasticities. The regression approach remains valid even when markets are not perfectly competitive as long as *quantities* (as opposed to value products) of inputs and output are known.[3]

To fix ideas, consider the running example from the main text, an economy that uses two natural capital inputs, N_1 and N_2, with output elasticities γ_1 and γ_2 (in addition to produced capital and labor). Suppose a national accountant observes time series data of change in output, g_Y, (i.e., change in real GDP) along with growth in produced capital g_K and labor g_L and measures all quantities she observes without error. Assume from here forward for all factors she measures it is done without measurement error. An econometrician who observes g_K and labor g_L has sufficient data to estimate γ_K and γ_L using a regression model based on the presupposed data generating process

[3] This condition is rarely met in practice as output is typically measured in terms of monetary value as opposed to physical quantities.

$$g_Y = \hat{g}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \tag{Model 1}$$

where $\hat{g}_A$ is an intercept term that becomes an estimator of average TFP growth. However, equation (3), implies that the true data generating process (DGP) is

$$g_Y = \gamma_K g_K + \gamma_L g_L + \gamma_1 g_{N_1} + \gamma_2 g_{N_2} + g_{TFP} \tag{DGP}$$

If one estimates (Model 1) by ordinary least squares (OLS), the fitted coefficients will be subject to traditional omitted variable bias (OVB) (Hansen). For example, when $\hat{\gamma}_K$ is the OLS estimand from Model 1 it identifies

$$\hat{\gamma}_K = \gamma_K + \gamma_1 \kappa_{N_1,\tilde{K}} + \gamma_2 \kappa_{N_2,\tilde{K}} + \kappa_{TFP,\tilde{K}} \tag{Bias 1}$$

where $\kappa_{X,\tilde{K}}$ is the coefficient from a linear projection of the growth of X onto the residualized growth in K. By (Bias 1) unless growth in capital, K, and in labor, L, are uncorrelated with growth in the omitted natural capital factor inputs and TFP, estimates of $\hat{\gamma}_K$ and $\hat{\gamma}_L$ from Model 1 will be biased. The magnitude of the bias depends on the magnitude of the output elasticities for omitted factors and on how they covary with observed factors.

Consider an alternative case where one of the productive natural capital stocks, N_1, is observed by the econometrician. The analyst may expect that including this additional factor will reduce the bias in the estimates. They consider estimating

$$g_Y = \tilde{g}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \tag{Model 2}$$

where tildes differentiate Model 2 from Model 1. The bias in the estimate of the capital output elasticity using (Model 2) is

$$\tilde{\gamma}_K = \gamma_K + \gamma_2 \lambda_{N_2,\tilde{K}} + \lambda_{TFP,\tilde{K}} \tag{Bias 2}$$

where the λs are coefficients from residualized auxiliary regressions as above. It is ambiguous whether (Bias 1) or (Bias 2) is greater without further assumptions.

Consider first the omitted variable bias when g_{N_1} and g_K are the only terms in the growth accounting equation that are not iid. Predictions of theory are immediate; OVB is present in model 1 only, and including g_{N_1} always (weakly) reduces bias when estimating γ_K. Now consider relaxing this assumptions slightly, such that $\rho(g_{N_1}, g_{TFP}) > 0$ while g_{TFP} remains uncorrelated with all other growth terms (including g_K).[4] Both models 1 and 2 now produce biased estimates of γ_K because $\kappa_{N_1,\tilde{K}}$ and $\lambda_{N_1,\tilde{K}}$ are nonzero, making it ambiguous which estimator is less biased. For Model 1, the bias is determined by $\kappa_{N_1,\tilde{K}}$, which is the correlation between g_{N_1} and the portion of the growth in capital that is not explained by growth in labor:

[4] The choice of direction for the inequality here is arbitrary and symmetric results would follow if it were reversed.

$$\hat{\gamma}_K - \gamma_K = \gamma_1 \kappa_{N_1,\tilde{K}} \tag{Bias 1'}$$

When the correlation between growth in N_1 and the growth in capital not explained by growth in other facts is positive ($\kappa_{N_1,\tilde{K}} > 0$), some of the growth in output caused by growth in N_1 is misattributed to produced capital, leading to an overestimate γ_K. In Model 2, the bias is given by:

$$\tilde{\gamma}_K - \gamma_K = \lambda_{A,\tilde{K}} \tag{Bias 2'}$$

and will lead to an RMSE that depends on the magnitude of the correlation between growth in N_1 and growth in TFP. The bias in (Bias 2′) has the opposite sign of the correlation between growth in N_1 and TFP because the regression coefficient $\lambda_{A,\tilde{K}}$ is the covariance in TFP and the portion of capital growth *not* explained by g_{N_1}. Since g_K and g_{TFP} are *unconditionally* uncorrelated, as g_{TFP} is positively correlated with g_{N_1} and a portion of g_K that is correlated with g_{N_1}, the bias results from the negatively correlated with the portion of g_k that's orthogonal to g_{N_1}, which leads to an overall correlation that is zero. The result is that g_{TFP} is negatively correlated with $\tilde{K}$, the residualized portion of g_K, so that γ_K is underestimated when $\rho(g_{N_1}, g_{TFP}) > 0$.

1.4 Estimating TFP Growth Using Residuals

We now turn to examining how biases in estimated output elasticities propagate through to the estimation of TFP. This section begins with a general case where take estimated output elasticities as given before turning to the specific case where they are estimated using the OLS approaches described in Section 1.3 above. From here forward, we operate under the assumption that TFP growth g_{TFP} is an independent and identically distributed random variable.

For a national accountant with perfect information on output growth, output elasticities, and factor growth inputs, her estimand of interest when calculating expected TFP growth would be

$$\mathbb{E}[g_{TFP}] = \mathbb{E}[g_Y] - \gamma_K \mathbb{E}[g_K] - \gamma_L \mathbb{E}[g_L] - \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \tag{6}$$

and simply plugging the sample analogs of population moments on the right hand side of (6) would be a feasible and unbiased estimator of $\mathbb{E}[g_{TFP}]$.[5] However, as discussed above and in the main text, it is often the case that measurement of natural capital is incomplete.

Turning again to our running example, suppose a natural accountant only has access to data on growth in output, produced capital, and labor inputs. One option for estimating TFP is to use a version of the estimator in Model 1 which omits both forms of natural capital but accounts for the two factors she measures. With estimated output elasticities $\{\hat{\gamma}_K, \hat{\gamma}_L\}$ in hand, her estimator for average TFP growth $\mathbb{E}[g_{TFP}]$ is

$$\hat{g}_A \triangleq \bar{g}_Y - \hat{\gamma}_K \bar{g}_K - \hat{\gamma}_L \bar{g}_L \tag{7}$$

[5]We assume all relevant moment conditions hold such that all estimands are well-defined.

where bars denote sample averages. Let

$$\varepsilon \triangleq \hat{g}_A - \mathbb{E}[g_{TFP}] \tag{8}$$

be the difference between (6) from (7). Under the assumption she measures growth in produced capital and labor without error, taking expectations of (8) conditional on estimated output elasticities gives the conditional bias of the estimator in (7)

$$\mathbb{E}[\varepsilon|\{\hat{\gamma}_K, \hat{\gamma}_L\}] = [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \tag{9}$$

gives the bias of the average TFP growth estimator in (7).[6] The shorthand ε will ease notation going forward. Define the mean squared error (MSE) for this estimator as $\text{MSE}(1) \triangleq \mathbb{E}[\varepsilon^2]$.

Now, suppose the national accountant is also able to measure g_{N_1} without error and separately estimates a new set of output elasticities $\{\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_{N_1}\}$. If we augment equation (7) with an additional term capturing the contribution of N_1 to growth, our alternative estimator based on Model 2 is

$$\tilde{g}_A \triangleq \bar{g}_Y - \tilde{\gamma}_K \bar{g}_K - \tilde{\gamma}_L \bar{g}_L - \tilde{\gamma}_1 \bar{g}_{N_1} \tag{10}$$

Letting $\delta \triangleq \tilde{g}_A - \mathbb{E}[g_{TFP}]$, the conditional bias for the estimator in Model 2 is

$$\mathbb{E}[\delta|\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_{N_1}] = \mathbb{E}[\gamma_K - \tilde{\gamma}_K]\mathbb{E}[g_K] + \mathbb{E}[\gamma_L - \tilde{\gamma}_L]\mathbb{E}[g_L] + \mathbb{E}[\gamma_1 - \tilde{\gamma}_1]\mathbb{E}[g_{N_1}] + \sum_{j=2}^{J} \gamma_j g_{N_j} \tag{11}$$

where δ is again a useful shorthand going forward. The mean squared error for this estimator is defined as $\text{MSE}(2) \triangleq \mathbb{E}[\delta^2]$.

The magnitudes of biases and mean squared errors for each estimator depend in principle on both the estimated output elasticities as well as expected values of the growth rates of inputs. At this stage, it is ambiguous which estimator performs better under either criteria. We turn now to examining under what additional assumptions we can sign the bias in estimated TFP, and under what conditions including the first natural capital stock on the margin—shifting from Model 1 to Model 2—improves both of these criteria.

Our first result is formed under assumption that the production function displays constant returns to scale (like in 2). In this case, the output elasticities must sum to one. This implies that estimates of output elasticities used for the estimator derived from Model 1 must satisfy the identity $\hat{\gamma}_L = 1 - \hat{\gamma}_K$, and for Model 2 must satisfy $\tilde{\gamma}_L = 1 - \tilde{\gamma}_K - \tilde{\gamma}_1$. We may then state the following:

[6] Note equation above implicitly assumed that the expectations of $\bar{g}_L$ and $\bar{g}_K$ are independent of estimated output elasticities which is unlikely to hold in practice.

Proposition 1. When production exhibits constant returns to scale, if $\min_j\{g_{N_j}\} > \max\{g_K, g_L\}$, then $\mathbb{E}[\varepsilon] > 0$, and TFP growth is overestimated. If $\max_j\{g_{N_j}\} < \min\{g_K, g_L\}$, then $\mathbb{E}[\varepsilon] < 0$ and TFP growth is underestimated. In either case, adding an additional natural capital stock N_j to the growth accounting equation unambiguously reduces bias as long as the econometrician estimates g_{N_j} and γ_j without bias. When neither case holds, the sign of the bias is ambiguous, even if all other elasticities and growth rates for included factor inputs are measured without error.

Proof. See Section 6.1. □

Even if the econometrician correctly imposes the assumption of CRS production and measures all factors and output elasticities without error, she cannot generally sign the bias induced by leaving out natural capital. Only under very restrictive assumptions on the growth rates of natural capital can she be sure of the sign of the bias and that inclusion of additional natural capital stocks is helpful on the margin. Otherwise, including an additional natural capital stock (going from model 1 to model 2) is not guaranteed to reduce the bias or mean squared error in her estimate even under very optimistic assumptions on measurement:

Lemma 1. Suppose an econometrician has perfect measurement of all growth rates and output elasticities for factor inputs she observes. Suppose further that she knows with certainty that growth rates for all factor inputs are independent and identically distributed. The effect of adding of an omitted natural capital stock to the growth accounting equation (7) is still ambiguous in terms of bias and mean squared error of the growth accounting-based TFP estimator unless it is the only remaining omitted natural capital stock.

Proof. See Section 6.2. □

In the case of perfect measurement of output elasticities and growth in factor use, the biases for TFP estimates from Model 1 and 2 are

$$\mathbb{E}[\varepsilon|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L)] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}]$$

$$\mathbb{E}[\delta|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1)] = \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j}]$$

respectively. As hinted at in the main text, only when all growth terms share the same sign in expectation is it guaranteed that $|\mathbb{E}[\delta]| \leq |\mathbb{E}[\varepsilon]|$. Without the all-the-same-signs condition the effect of adding natural capital stocks to (7) yields ambiguous effect on bias and MSE.

An alternative formulation for the bias from these estimators is to express the omitted as deviations from average growth across all natural capital factors that are omitted from a given model. Let $\mathcal{J}$ be the set of natural capital factors omitted under Model 1. The average factor shares and growth rates across these factors are

$$\bar{\gamma}^{\mathcal{J}} \triangleq \frac{1}{|\mathcal{J}|}\sum_{j\in\mathcal{J}}\gamma_j \quad , \quad \bar{g}^{\mathcal{J}} \triangleq \frac{1}{|\mathcal{J}|}\sum_{j\in\mathcal{J}}\mathbb{E}[g_{N_j}]$$

such that for Model 1 the bias is

$$\mathbb{E}[\varepsilon|\mathcal{J}] = J\mathbb{E}\left[\gamma_j\mathbb{E}[g_{N_j}]|\mathcal{J}\right] = J\left(\mathrm{Cov}\Big(\gamma_j, \mathbb{E}[g_j]\Big|j\in\mathcal{J}\Big) + \bar{\gamma}^{\mathcal{J}}\bar{g}^{\mathcal{J}}\right) \tag{12}$$

and for Model 2, where included factors are set $\mathcal{J}'$ with cardinality $J-1$ the bias is

$$\mathbb{E}[\delta|\mathcal{J}'] = (J-1)\mathbb{E}\left[\gamma_j\mathbb{E}[g_{N_j}]|\mathcal{J}'\right] = (J-1)\left(\mathrm{Cov}\Big(\gamma_j, \mathbb{E}[g_j]\Big|j\in\mathcal{J}'\Big) + \bar{\gamma}^{\mathcal{J}'}\bar{g}^{\mathcal{J}'}\right) \tag{13}$$

Proposition 2. Suppose that the variances of γ_j and $\mathbb{E}[g_j]$ are sufficiently small so that for any subset of natural capital stocks $\mathcal{J}'$ with cardinality $J-1$ the inequality

$$\frac{J}{J-1} \geq \frac{\bar{g}^{\mathcal{J}'}\bar{\gamma}^{\mathcal{J}'}}{\bar{g}^{\mathcal{J}}\bar{\gamma}^{\mathcal{J}}}$$

holds. Then when the covariance between γ_j and $\mathbb{E}[g_{N_j}]$ across capital stocks is zero, adding a correctly-measured capital stock and its output elasticity to the growth accounting equation lowers the magnitude of bias.

Proof. See Section 6.3. □

The assumptions on sign monotonicity required to ensure including a natural capital stock on the margin reduces bias can thus be weakened slightly in exchange for the assumption that the growth rate of natural capital inputs is uncorrelated with their output elasticities. For this to hold, individual natural capital stocks must grow in a similar manner to the ensemble average over time, the growth rates must not be too strongly correlated with their factor shares, and output elasticities and factor growth rates must be estimated without error. In practice, we view that even these weaker assumptions are unlikely to hold in the data.

2 Regression Methodology and Analysis

This section describes how we implement the regression model in Section 1 of the main paper that estimates the relationship between measured TFP growth and changes in natural capital stocks and at the country level.

2.1 Theory

When a national accountant measures productivity growth using the Solow-residual based method described in equation 3 of the main text, conditional on her choice of inputs $\boldsymbol{I}$ the estimator for TFP growth each period is

$$g_A(t)|\boldsymbol{I} = g_Y(t) - \sum_{i\in\boldsymbol{I}} \hat{\gamma}_i \hat{g}_i(t) = g_{TFP}(t) \ + \sum_{x\in\boldsymbol{X}} \gamma_x g_x(t) \ - \sum_{i\in\boldsymbol{I}} \hat{\gamma}_i \hat{g}_i(t) \tag{14}$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are her estimates of the growth rates and output elasticities of included inputs. When the Solow-residual based method is correctly specified such that $\boldsymbol{I} = \boldsymbol{X}$, (14) simplifies to

$$g_A(t)|\boldsymbol{I} = g_{TFP}(t) \ + \sum_{x\in\boldsymbol{X}} \gamma_x g_x(t) - \hat{\gamma}_x \hat{g}_x(t) \tag{15}$$

In our case in Section 1, there are two reasons to expect the second term on the RHS to be nonzero. First, even if we were certain we knew the seven renewable capital stocks measured in the CWON were the only ones contributing to production, we are still working with *estimated* growth rates $\hat{g}_x$ as opposed to observations without error. Second, the same logic applied to the underlying construction process for the factors that are included when a given measured series for TFP is constructed; the estimates of output elasticities and factor growth rates $\hat{\gamma}_x$ and $\bar{g}.$ that enter the PWT's growth accounting equation may also be measured with error.

This leads to the decomposition of the series we present in Section 1 of the main text. The specific TFP growth series we use from the Penn World Tables (PWT) version 10.01 (Feenstra, Inklaar, and Timmer)—$RTFP^{NA}$, uses a version of (14) where natural capital services are not included in the set of included inputs $\boldsymbol{I}$. Assuming that the specification in the PWT captures all other factor inputs outside of natural capital (i.e., where the set $\boldsymbol{X} \setminus \boldsymbol{I}$ is comprised solely of natural capital inputs), (14) takes the form

$$g_A(t)|\boldsymbol{I}, \hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} \ + \underbrace{\sum_{x\in\boldsymbol{X}\setminus\boldsymbol{I}} \gamma_x g_x(t)}_{\text{Omitted inputs}} \ + \underbrace{\sum_{i\in\boldsymbol{I}} [\gamma_i g_i(t) \ - \ \hat{\gamma}_i \hat{g}_i(t)]}_{\text{Measurement error in included inputs}} \tag{16}$$

Thus measured TFP growth at the country level (from the Penn World Tables or other sources) can be correlated with growth in a given natural capital stock in that country—g_{N_i}—through three channels: (1), correlation between the growth of true TFP (innovation, technical change) and changes in the use of N as an input; (2), resource N_i is an omitted productive input that is absorbed when g_A is constructed as a residual and (3), because of correlations between natural capital use and any measurement error in resources that are included in the original set of inputs $\boldsymbol{I}$ when g_A is constructed.

We wish to test the null hypothesis for channel (2)—that N_i is not an omitted variable—in isolation. As discussed in the main text, because we only observe g_A jointly as the sum of the three terms in (16), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. In practice, we test (and reject) a joint null hypothesis that a proposed set of seven omitted natural capital stocks is uncorrelated with measured TFP growth g_A. Thus even when our results are statistically robust, it is possible that they are driven by underlying relationships between natural capital and TFP growth or measurement error as opposed to being omitted variables in the growth accounting equation. However, we believe such a result suggests that it is desirable to collect and organize the data systematically to resolve this question.

2.2 Data for Covered Countries in the Empirical Portion of the Main Text

We estimate equation (4) from the main text on a balanced panel of 117 countries over the 1996-2019 period. Table 1 displays the list of countries covered in the regression.

Table 1: Covered Countries

Angola	Argentina	Armenia
Australia	Austria	Bahrain
Barbados	Belgium	Benin
Bolivia	Botswana	Brazil
Bulgaria	Burkina Faso	Burundi
Cameroon	Canada	Central African Republic
Chile	China	Colombia
Costa Rica	Cote d'Ivoire	Croatia
Cyprus	Czech Republic	Denmark
Dominican Republic	Ecuador	Egypt
Estonia	Eswatini	Fiji
Finland	France	Gabon
Germany	Greece	Guatemala
Honduras	Hong Kong	Hungary
Iceland	India	Indonesia
Iran	Iraq	Ireland
Israel	Italy	Jamaica
Japan	Jordan	Kazakhstan
Kenya	South Korea	Kuwait
Kyrgyzstan	Laos	Latvia
Lesotho	Lithuania	Luxembourg
Macao	Malaysia	Malta
Mauritania	Mauritius	Mexico
Moldova	Mongolia	Morocco
Mozambique	Namibia	Netherlands
New Zealand	Nicaragua	Niger
Nigeria	Norway	Panama
Paraguay	Peru	Philippines
Poland	Portugal	Qatar
Romania	Russian Federation	Rwanda
Saudi Arabia	Senegal	Serbia
Sierra Leone	Singapore	Slovakia
Slovenia	South Africa	Spain
Sri Lanka	Sudan	Sweden
Switzerland	Tajikistan	Tanzania
Thailand	Togo	Trinidad and Tobago
Tunisia	Türkiye	Ukraine
United Kingdom	United States	Uruguay
Venezuela	Zambia	Zimbabwe

Table 2 shows the seven renewable capital stocks in the CWON that form our set of environmental inputs $\boldsymbol{I}$ along with the native quantity units of measurement and the coverage of the underlying data. The urban land area variable in the CWON dataset is taken from the Center for International Earth Science Information Network (CIESIN) and Columbia University. The timber land variable is constructed based on the total hectares of productive timber area at the country level as measured by the Food and Agriculture Organization of the United Nations (FAO). The agricultural land variable is constructed using data from the World Bank's World Development Indicators series based on total crop and pasture area measured by the FAO (World Bank). Non-timber forest cover measures the total area of forests providing ecosystem services beyond wood products and is intended to capture the change in "... the broader ecological contributions of forests, including carbon storage, water regulation, and the provision of non-wood forest products (NWFP)" over time (World Bank). CWON data on mangrove ecosystem coverage are taken from the Global Mangrove Watch (1996 - 2020) Version 3.0 Dataset (Bunting et al.). Data on annual fisheries biomass were provided to the authors of the CWON by researches at the Sea Around Us research initiative at the University of British Columbia (Pauly, Zeller, and Palomares). Finally, the CWON data on annual hydroelectric power generation are attributed to Midsummer Analytics (World Bank).

Due to the paucity of data spanning the entire panel for some resources, we impute zeros to the growth rates for all resources where the observation in the prior year is a missing value. This imposes the assumption that there is no change in the true usage rate of natural services where data from the year prior is missing. The last column shows the share of data points that are imputed values of zero for each resource. Outside of mangroves, biomass, and hydropower, the imputation is minimal; however, for the latter three resources a huge portion of the original data are missing entries. To ensure our results are not driven by the imputation process, Table 4 shows that our results hold when we remove resources where a large fraction of observations are imputed and re-estimate equation 4 in the main text.

Table 2: Renewable resources used to estimate equation 4 in the main text

Resource	**Unit**	**% Obs. Missing**	**# Countries**
Urban land area	Index	0.00%	117
Productive timber forest area	Hectares	1.25%	114
Agricultural land area	Square Kilometers	2.56%	116
Non-timber forest cover	Square Kilometers	3.95%	113
Mangrove ecosystem area	Hectares	63.25%	43
Fishery Biomass	Metric tons	23.08%	90
Hydroelectric Power Generation	Gigawatt-hours	27.07%	86

2.3 Robustness Checks

We turn here to presenting a number of robustness checks of our empirical exercise in the main text. While we cannot test whether our results are driven by correlation between natural capital stock growth and objects (1) and (3) in equation (16), this section ensures that the results are not driven by our choice of outcome variables or included natural capital stocks.

2.3.1 Randomization Inference Tests of Joint Significance

While our estimating sample is a balanced panel, it's still possible that individual operations with high leverage are creating spurious results of joint significance. To check for this we perform a randomization inference test on the specification in column (3) of Table 1 in the main text. To operationalize this test, we resample growth in the seven natural capital stocks, $g_{N_{n,j,t}}$, within each country without replacement and randomly reassign them to years within the panel. This process, by construction, imposes that the correlation between these newly reassigned regressors $\tilde{g}_{N_{n,j,t}}$ and the outcome variable $g_{A_{j,t}}$ is expected to be zero within each country. We repeat this process 100,000 times. For each permuted dataset, we estimate the following regression

$$g_{A_{j,t}} = \sum_{i \in \boldsymbol{I}} \beta_i \tilde{g}_{N_{i,j,t}} + \sum_{z \in \boldsymbol{Z}} \theta_z g_{z,j,t} + \delta_j + \varepsilon_{j,t} \tag{17}$$

and perform a Wald test of joint significance for coefficients $\boldsymbol{\beta}$ each time. Because the data are constructed such that the null hypothesis for the joint Wald test is true, this test forms an empirical distribution for $F-$test statistics under the null hypothesis. Figure 1 shows the empirical distribution of test statistics when the null statistic is true for 100,000 resampled versions of our panel. The p value from this test–the share of placebo F test statistics that are larger than the one in our estimating sample–is 0.015.

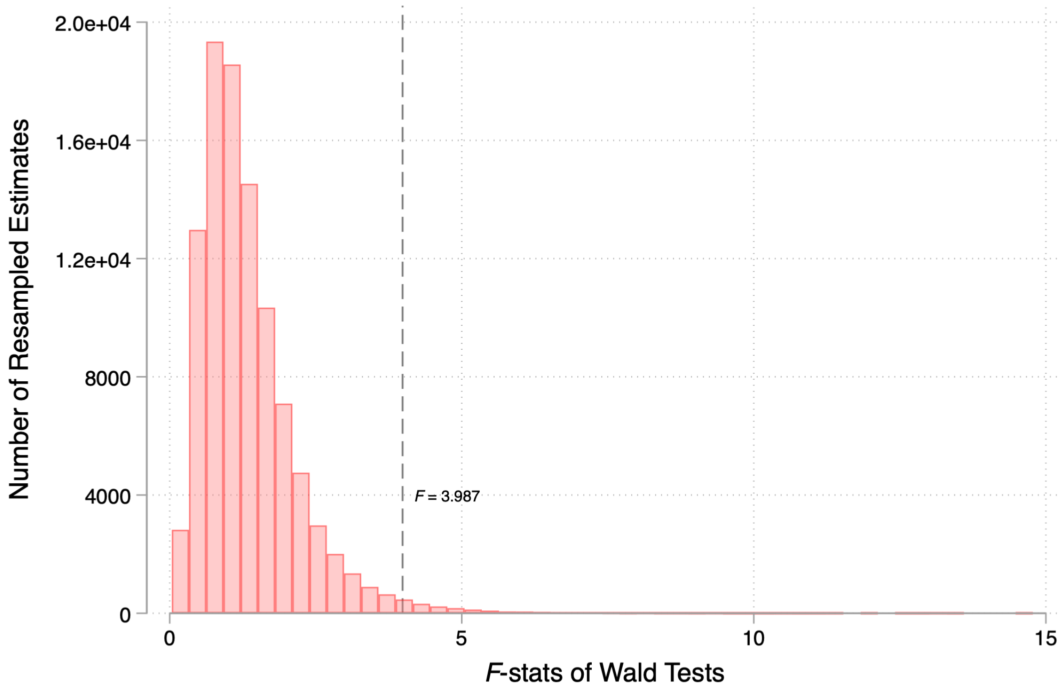

Figure 1: Histogram of F-statistics from Wald tests of joint nullity performed on 100,000 permuted samples of the data where the null is imposed true in expectation.

2.3.2 Choice of Covariates

The seven natural capital stocks in Table 2 comprise the baseline set we test because these are the seven in the CWON dataset for which quantity data are public.[7] However, as the choice to include all seven is still somewhat arbitrary given they are ultimately only a subset of renewable natural capital, here we ensure that our rejection of the joint null result is not driven by our choice of regressors.

Excluding Urban Land Table 3 displays the results when equation 4 in the main text is re-estimated excluding urban land from the set of renewable natural capital stocks. The vast majority of estimated coefficients remain positive for natural capital stocks excluding forest cover, in line with our results in the main text. Wald tests of joint nullity using asymptotic standard errors and semi-parametric methods lose some precision, but p-values still hover around traditional thresholds for statistical significance.

[7]Data used to construct fisheries wealth in the CWON dataset are not made public.

Table 3: TFP Growth vs. Natural Capital Growth Excluding Urban Land

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Timber Land	0.164	0.166	0.071	0.075	-0.003	0.026
	(0.090)	(0.075)	(0.074)	(0.071)	(0.068)	(0.104)
Agricultural Land	0.027	0.043	0.030	0.036	0.041	0.043
	(0.025)	(0.025)	(0.022)	(0.023)	(0.021)	(0.017)
Forest Cover	-0.067	-0.111	-0.005	0.012	0.083	-0.027
	(0.094)	(0.091)	(0.079)	(0.083)	(0.053)	(0.076)
Mangroves	0.122	0.067	0.357	0.327	0.318	-0.108
	(0.244)	(0.242)	(0.299)	(0.322)	(0.367)	(0.427)
Fishery Biomass	0.112	0.026	0.094	0.072	0.047	-0.104
	(0.098)	(0.077)	(0.108)	(0.101)	(0.135)	(0.108)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.063	0.036	0.109	0.074	0.059	0.045
Bootstrap Wald	0.094	0.070	0.125	0.078	–	0.086
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Exclusion of Resources with Large Shares of Missing Observations Table 4 displays the results when equation 4 in the main text is re-estimated on a subset of four natural capital stocks excluding those where a large share of observations are either imputed or interpolated (see Table

2). We remain able to reject the joint nullity of estimated coefficients using asymptotic standard errors and semi-parametric methods at traditional levels.

Table 4: TFP Growth vs. Natural Capital Growth, Excluding Stocks with Large Portions of Missing Data

Outcome: $\Delta\ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.295	0.298	0.198	0.209	0.146	0.062
	(0.092)	(0.096)	(0.065)	(0.073)	(0.061)	(0.115)
Timber Land	0.171	0.171	0.078	0.082	0.001	0.017
	(0.093)	(0.076)	(0.075)	(0.072)	(0.068)	(0.107)
Agricultural Land	0.029	0.045	0.029	0.035	0.041	0.046
	(0.025)	(0.025)	(0.022)	(0.022)	(0.021)	(0.017)
Forest Cover	-0.069	-0.114	-0.003	0.013	0.080	-0.028
	(0.097)	(0.092)	(0.079)	(0.083)	(0.053)	(0.076)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.063	0.036	0.109	0.074	0.059	0.045
Bootstrap Wald	0.094	0.070	0.125	0.078	–	0.086
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta\ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Excluding Urban, Agriculture, and Timber Land Table 5 displays the results when equation 4 in the main text is re-estimated excluding all forms of land resources outside of forests from the set of renewable natural capital stocks. This will avoid any bias induced by land resources if they are correlated with measurement error in capital or labor income in national accounts that is used to estimate TFP growth in the PWT. Wald tests of joint nullity using asymptotic standard errors and semi-parametric methods become marginal or insignificant at around traditional thresholds for statistical significance under some specifications we consider. As discussed when estimating the specification in Table 4, some of this loss in precision may be due to the large share of missing values for factor growth we impute as zeros here.

Table 5: TFP Growth vs. Natural Capital Growth, Excluding Land

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Forest Cover	0.068	0.027	0.047	0.067	0.086	-0.013
	(0.061)	(0.062)	(0.068)	(0.075)	(0.043)	(0.070)
Mangroves	0.138	0.085	0.348	0.317	0.303	-0.096
	(0.240)	(0.237)	(0.298)	(0.322)	(0.366)	(0.428)
Biomass	0.114	0.031	0.092	0.070	0.039	-0.109
	(0.098)	(0.078)	(0.107)	(0.101)	(0.134)	(0.107)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.050	0.061	0.111	0.107	0.062	0.229
Bootstrap Wald	0.067	0.108	0.128	0.116	–	0.185
# Observations	2,808	2,808	2,808	2,808	2,808	2,457

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Inclusion of Non-Renewables The analysis in the main text looked at the correlation between TFP growth and changes in stocks of renewable resources as proxies for changes in services from natural capital. It is possible that the significant results there were due to omitted variable bias from leaving out non-renewable resources. Table 6 repeats the exercise in table 1 of the main text when we include all natural capital flows that can be calculated from the public version of the CWON dataset (i.e., renewable, fossil, and subsurface minerals). The full set of non-renewable and renewable natural capital stocks included are listed in Tables 11 and 12. Table 6 shows the estimated output elasticities associated with each natural capital stock when equation 4 in the main text is estimated on an unbalanced panel of 118 countries between 1972 and 2019 after expanding $\boldsymbol{I}$ to the full set of 20 natural capital stocks. Table 7 re-estimates this specification for the 1996-2019 period so it coincides with the time coverage in the main estimating sample.[8]

Overall, including the full suite of natural capital covariates available in the CWON dataset does not change the qualitative aspect of our results. Reassuringly, coefficients on oil and natural gas (resources where the associated industries comprise a substantial share of global GDP) are positive and significant at traditional levels in both tables. Urban land and hydropower remain precisely estimated. More generally, the coefficients on most natural capital stocks remain positive, in line with our priors that the use of non-renewable resources in production should be "goods" in terms of increasing output. While the coefficient on forest cover is negative, it is not significant at traditional levels and qualitatively aligned with some specifications shown in the main text. For these specifications including a broader suit of natural capital, we the null hypothesis to include all 20 natural capital stocks. Wald tests using both asymptotic and semi-parametric methods become substantially tighter around zero in all specifications we consider.

[8]The resulting number of observations and covered countries in this appendix does not coincide with that in the main text as here we adopt a more aggressive imputation strategy in order to fill in missing data. The process for constructing these quantity changes prior to 1996 is described in Section 3. Results are qualitatively unchanged when the estimating sample here is restricted to the 1996-2019 period, as shown in Table 7.

Table 6: TFP Growth vs. Renewable and Non-Renewable Natural Capital Growth

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Bauxite	0.002	0.002	0.001	0.001	0.000	-0.001
	(0.003)	(0.004)	(0.003)	(0.004)	(0.003)	(0.003)
Coal	0.005	0.007	0.009	0.009	0.007	0.004
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Oil	0.014	0.014	0.016	0.015	0.015	0.015
	(0.007)	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)
Copper	0.000	0.001	0.002	0.002	0.002	-0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Phosphate	0.014	0.014	0.015	0.015	0.015	0.017
	(0.008)	(0.008)	(0.009)	(0.008)	(0.009)	(0.011)
Natural Gas	0.002	0.003	0.003	0.003	0.003	0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
Gold	0.002	0.002	0.003	0.003	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Silver	0.001	0.001	0.002	0.002	0.002	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Iron	0.002	0.003	0.001	0.002	0.003	0.004
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Tin	0.002	0.002	0.004	0.004	0.004	-0.000
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Lead	0.001	0.001	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Zinc	-0.000	-0.000	-0.001	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Nickel	0.003	0.003	0.002	0.002	0.002	0.001
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Urban Land	0.294	0.307	0.225	0.239	0.249	0.099
	(0.080)	(0.093)	(0.061)	(0.068)	(0.074)	(0.078)
Agricultural Land	0.201	0.192	0.110	0.109	0.076	0.143
	(0.102)	(0.076)	(0.085)	(0.084)	(0.067)	(0.098)
Timber Land	0.015	0.034	0.037	0.046	0.042	0.037
	(0.025)	(0.025)	(0.023)	(0.023)	(0.021)	(0.018)
Forest Cover	-0.107	-0.171	-0.178	-0.206	-0.081	-0.084
	(0.105)	(0.091)	(0.095)	(0.095)	(0.090)	(0.104)
Mangroves	0.094	0.099	0.149	0.172	0.333	0.157
	(0.241)	(0.218)	(0.259)	(0.257)	(0.211)	(0.302)
Fishery Biomass	-0.101	-0.120	-0.020	-0.026	0.171	0.021
	(0.069)	(0.058)	(0.074)	(0.074)	(0.106)	(0.103)
Hydropower	0.013	0.013	0.012	0.012	0.010	0.010
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)
Wald	0.002	0.000	0.001	0.000	0.000	0.014
Bootstrap Wald	0.096	0.044	0.032	0.014	–	0.111
# Observations	5,255	5,255	5,255	5,255	5,255	4,901
# Countries	118	118	118	118	118	118

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 7: TFP Growth vs. All NK Growth, 1996-2019 Sample

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Bauxite	0.003	0.003	0.001	0.001	-0.001	-0.001
	(0.006)	(0.006)	(0.005)	(0.005)	(0.004)	(0.003)
Coal	0.004	0.006	0.002	0.003	0.003	0.001
	(0.006)	(0.005)	(0.004)	(0.004)	(0.005)	(0.004)
Oil	0.003	0.003	0.003	0.002	0.002	0.003
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
Copper	-0.003	-0.003	-0.003	-0.003	-0.004	-0.007
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
Phosphate	0.018	0.017	0.021	0.020	0.020	0.020
	(0.011)	(0.011)	(0.013)	(0.013)	(0.014)	(0.013)
Natural Gas	0.000	0.000	-0.000	-0.000	-0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Gold	0.003	0.002	0.002	0.002	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Silver	0.002	0.002	0.003	0.003	0.003	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Iron	0.004	0.004	0.004	0.004	0.005	0.007
	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Tin	-0.002	-0.001	-0.001	-0.001	-0.001	0.001
	(0.002)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)
Lead	0.002	0.002	0.002	0.002	0.002	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Zinc	-0.002	-0.002	-0.003	-0.003	-0.003	-0.004
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.003)
Nickel	0.003	0.002	0.000	0.000	-0.000	-0.001
	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.003)
Urban Land	0.311	0.311	0.209	0.219	0.152	0.170
	(0.092)	(0.097)	(0.065)	(0.073)	(0.060)	(0.064)
Agricultural Land	0.157	0.159	0.069	0.074	-0.009	0.147
	(0.090)	(0.076)	(0.073)	(0.071)	(0.065)	(0.089)
Timber Land	0.028	0.044	0.032	0.038	0.044	0.036
	(0.024)	(0.025)	(0.021)	(0.021)	(0.021)	(0.017)
Forest Cover	-0.067	-0.109	-0.014	0.004	0.083	0.075
	(0.095)	(0.091)	(0.081)	(0.083)	(0.053)	(0.103)
Mangroves	0.147	0.091	0.357	0.327	0.315	0.166
	(0.243)	(0.239)	(0.281)	(0.304)	(0.337)	(0.286)
Fishery Biomass	0.120	0.035	0.122	0.099	0.068	0.052
	(0.094)	(0.074)	(0.111)	(0.104)	(0.129)	(0.107)
Hydropower	0.011	0.012	0.010	0.010	0.008	0.010
	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
Wald	0.016	0.014	0.023	0.028	0.021	0.001
Bootstrap Wald	0.096	0.044	0.032	0.014	–	0.111
# Observations	2832	2832	2832	2832	2832	2802
# Countries	118	118	118	118	118	118

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for . . . " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

2.3.3 Alternative Productivity Measures

This section examines whether we find similar correlations between natural capital and productivity growth when we consider alternative measurements of TFP beyond those in the Penn World Tables. We consider two panels here: one from Eurostat covering the 27 countries in the European (plus Norway) which measures economy-wide multi-factor productivity (European Commission and Eurostat), and a second covering only the agricultural sector produced by the Economic Research Service at the USDA (Fuglie).

Eurostat MFP Table 8 displays the results when equation 4 in the main text is re-estimated on multifactor productivity series generated by the European Statistical Agency (Eurostat) for the 27 EU countries along with Norway (European Commission and Eurostat). The panel covers the years 2003-2019. Results from this alternative specification are shown in Table 8. While Wald tests of joint nullity using asymptotic standard errors remain significant, the semiparametric tests appear to lose some power.

Table 8: EU27 + Norway Eurostat MFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(MFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Urban Land	0.524	-0.010	0.266	0.027	-0.134
	(0.498)	(0.369)	(0.495)	(0.397)	(0.481)
Timber Land	-0.477	-0.711	-0.818	-0.815	-1.147
	(0.363)	(0.328)	(0.262)	(0.274)	(0.475)
Agricultural Land	0.027	0.030	0.035	0.041	0.035
	(0.040)	(0.034)	(0.037)	(0.035)	(0.033)
Forest Cover	0.308	0.328	0.421	0.408	0.418
	(0.234)	(0.257)	(0.175)	(0.189)	(0.260)
Fishery Biomass	-0.443	-0.522	0.268	0.198	0.243
	(0.148)	(0.172)	(0.190)	(0.199)	(0.521)
Hydropower	-0.002	0.001	0.001	0.001	0.002
	(0.003)	(0.002)	(0.003)	(0.002)	(0.002)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.006	0.001	0.007	0.001	0.010
Bootstrap Wald	0.283	0.052	0.168	0.057	–
# Observations	476	476	476	476	476
# Countries	28	28	28	28	28

Note: All independent variables in the table are measured in terms of log changes. The mangroves account is zero for all EU countries and is removed as a covariate for regressions in this table. Controls rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

The results for the sign and magnitude of coefficients when estimated on EU MFP data differ substantially from the estimates using the full panel of TFP from the Penn World Tables in the main text. The coefficients on timber and forest cover swap signs; coefficients on timber land also become statistically significant at traditional levels. In contrast with this, urban land loses its statistical significance. To examine whether this difference in estimated coefficients is driven by heterogeneity across geographies within the main sample, we re-estimate our main specification on a subsample restricted to subset of EU countries plus Norway. This "apples-to-apples" comparison re-runs the regression in Table 8 using the Penn World Tables TFP series as opposed to the Eurostat MFP series as an outcome variable. To ensure a fair comparison, we restrict the sample to the time truncated time period of 2003-2019. During this period, the Pearson coefficient between the two measures of TFP growth is $\rho = 0.81$

Results for comparison are shown in Table 9. Inspection of the results when the outcome for the EU27 plus Norway is re-estimated using PWT data yields qualitatively similar coefficient changes. The signs on forests and timber land also flip. However, while the tables pass an eye test, the joint estimates of the coefficients are statistically different. When we estimate the specification in column (4) jointly for each outcome variable using a seemingly unrelated regression, a Wald test of the equality of all coefficients on natural capital between the two models rejects the null hypothesis of equality far beyond standard levels ($p < 0.001$).

Table 9: EU27 + Norway PWT TFP Growth vs. Natural Capital Growth Growth

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Urban Land	0.787	0.349	0.474	0.312	0.265
	(0.598)	(0.372)	(0.506)	(0.373)	(0.427)
Timber Land	-0.218	-0.306	-0.165	-0.148	-0.449
	(0.186)	(0.189)	(0.090)	(0.129)	(0.277)
Agricultural Land	0.008	0.024	0.002	0.015	0.010
	(0.037)	(0.032)	(0.033)	(0.028)	(0.029)
Forest Cover	0.191	0.192	0.172	0.174	0.103
	(0.147)	(0.161)	(0.069)	(0.080)	(0.168)
Biomass	-0.275	-0.401	0.023	0.052	0.019
	(0.120)	(0.159)	(0.182)	(0.164)	(0.405)
Hydropower	-0.004	-0.000	0.000	0.001	0.001
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.006	0.013	0.264	0.436	0.106
Bootstrap Wald	0.442	0.299	0.920	0.974	–
# Observations	476	476	476	476	476
# Countries	28	28	28	28	28

Note: All independent variables in the table are measured in terms of log changes. The mangroves account is zero for all EU countries and is removed as a covariate for regressions in this table. Controls rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

USDA ERS Agricultural TFP Finally, we examine whether the renewable capital stocks we consider are correlated with a measure of TFP in the agriculture sector constructed by the USDA's Economic Research Service (Fuglie). The TFP measure from Fuglie differs qualitatively from measures in the PWT in that it explicitly accounts for one of the natural capital inputs that we consider, agricultural land, as well as more specialized inputs such as fertilizer use and tillage.

As such, we do not expect to see growth in agricultural land as a good predictor when regressing this measure of agricultural TFP growth on the natural capital stocks.

Table 10 shows results when equation 4 in the main text is re-estimated on a panel of agricultural TFP growth for 114 (163) countries between 1997 and 2016 depending on the availability of controls in the PWT. The coefficient on agricultural land area is statistically indistinguishable from zero, as expected given a different metric of agricultural land use is subtracted as an input when the outcome of $\Delta \ln(Ag.\ TFP_{j,t})$ is constructed (Fuglie). Tests of the joint null hypothesis performed using asymptotic and semiparametric approaches are marginal, and coefficients on individual resources fluctuate substantially depending on the specification we consider.

Table 10: Agricultural TFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(Ag.\, TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Urban Land	0.301	0.120	0.146	0.078	0.093
	(0.095)	(0.067)	(0.124)	(0.067)	(0.065)
Timber Land	0.164	-0.131	-0.067	-0.284	-0.374
	(0.090)	(0.204)	(0.123)	(0.312)	(0.454)
Agricultural Land	0.027	0.037	0.059	0.039	0.057
	(0.025)	(0.054)	(0.038)	(0.060)	(0.065)
Forest Cover	-0.067	0.157	0.078	0.221	0.276
	(0.094)	(0.189)	(0.121)	(0.255)	(0.308)
Mangroves	0.122	-0.108	0.247	0.426	0.736
	(0.244)	(0.167)	(0.156)	(0.172)	(0.392)
Biomass	0.112	0.072	-0.024	-0.055	0.005
	(0.098)	(0.100)	(0.106)	(0.122)	(0.204)
Hydropower	0.011	-0.012	-0.007	-0.015	-0.016
	(0.004)	(0.008)	(0.008)	(0.009)	(0.009)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.007	0.269	0.376	0.047	0.089
Bootstrap Wald	0.089	0.577	0.367	0.108	–
# Observations	2,808	2,377	3,399	2,377	2,377
# Countries	117	114	163	114	114

Note: All independent variables in the table are measured in terms of log changes. Control rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

3 Producing Törnqvist Quantity Indices for Natural Capital Aggregates

The theory and simulations in the main text consider how the marginal addition of one natural capital to the growth accounting equatoin affects estimates in two stock world. While this in theory corresponds to the case of renewable and non-renewable resources, in reality there multiple forms of natural capital under each of these classifications. Aggregating various non-renewable and renewable resources up to a single figure for each requires the construction of index numbers; summary statistics for the growth in the quantity of all renewable and non-renewable natural capital inputs used in production each year. This section describes the construction of Törnqvist quantity indices for renewable and non-renewable natural capital used to calibrate the simulations in Section 3 of the main text.

To aggregate the sets of renewable and non-renewable resources to single quantities, we form Törnqvist (Törnqvist et al.) quantity indices for each resource. The Törnqvist quantity index for renewable (non-renewable) resources is a harmonic mean over growth in each renewable (non-renewable) inputs that period, with each input weighted by its share of the total value of the full set of renewable (non-renewable) natural capital inputs. The process for constructing Törnqvist indices consists of three steps: (1) constructing quantities used in production and associated rents accruing to each resource in a given year, (2) computing value share weights, and (3) aggregating weighted growth into composite indices. It aligns closely with the procedure used by the World Bank when constructing the official *Changing Wealth of Nations* (CWON) wealth measures (World Bank), but performs all calculations excluding resources where data on quantities extracted or rents are proprietary and not made public.

3.1 Natural Capital Use and Rent Data

Data on quantities of resources extracted (for the case of non-renewables) and levels of stocks (for the case of renewables) are drawn directly from the World Bank CWON dataset (World Bank). Table 11 shows the 13 non-renewable natural capital stocks that are inputs into the non-renewable Törnqvist quantity index. Non-renewables consist of the three main fossil fuels that comprise roughly four-fifths of global energy use (i.e., coal, oil, and gas), most of the common ferrous and base metals, and a few precious metals used in production.

Table 11: Non-renewable resources used to construct empirical N_1

Resource	**Unit**
Coal	Metric tons
Gas	Terajoules[9]
Oil	Barrels
Bauxite	Metric tons
Copper	Metric tons
Gold	Metric tons
Iron	Metric tons
Lead	Metric tons
Nickel	Metric tons
Phosphate	Metric tons
Silver	Metric tons
Tin	Metric tons
Zinc	Metric tons

Table 12 shows the seven renewable natural capital stocks that are inputs into the renewable Törnqvist quantity index. These resources align directly with those used as independent variables in the regression analysis listed in Table 2.

Table 12: Renewable resources used to construct empirical N_2

Resource	**Unit**
Urban land area	Index
Productive timber forest area	Square Kilometers
Agricultural land area	Square Kilometers
Non-timber forest cover	Square Kilometers
Mangrove ecosystem Area	Hectares
Fishery biomass	Metric tons
Hydroelectric Power Generation	Gigawatt-hours

3.2 Data Construction

For each non-renewable resource, quantities and rents (i.e., income attributable to that resource in that year) are taken directly from the CWON dataset at the country-by-year level. Quantities of each resource r extracted in country j in year t are denoted as

$$Q_{j,t}^{(r)}$$

and are based on physical quantities (e.g., barrels of oil produced or tons of minerals mined). This approach considers only the use of non-renewable resources as inputs to extractive industries, where resources in their raw form are transformed into commodities sold as intermediate or final goods. This measure of non-renewable inputs differs substantially from the case where they are intermediate goods in production (e.g., crude oil as an input to oil refining), which are excluded when calculating GDP to avoid double counting. The associated rents are denoted

$$R^{(r)}_{j,t}$$

and are calculated by the World Bank based on on quantities extracted multiplied by local or global commodity market prices (World Bank).

For the case of renewables, quantities are constructed in the same manner, but based on physical measurement of stocks, rather than the flow of services provided. As discussed in the main text, this requires assuming that the services renewable natural capital stocks provide are done directly in proportion to physical quantities. Specifically, we require that services provided per unit of stock of a given resource are the same across countries and years. This lets us set the percent changes in renewable service flows each year equal to the percent changes in stocks, an object that is measured in the CWON data. We assign annual rents for renewables based on the flow value imputed to each service in the CWON data. These values are based on various studies that use direct or non-market valuation methods to assess the annual benefits from physical quantities of natural capital stocks. For example, the value of farmland is assigned based on the output of the agriculture it facilitates, while mangroves are valued based on the flood protection they provide (World Bank). All resource datasets are merged into a single country-by-year panel covering non-renewable (e.g., fossil fuels, minerals) and renewable (e.g., forests, agricultural land, hydropower) quantities and rents.

3.3 Gross Growth Rates

For each series from the CWON giving the quantity of a given resource used in country j, the gross growth rate is computed as:

$$\hat{Q}^{(r)}_{j,t} = \frac{Q^{(r)}_{j,t}}{Q^{(r)}_{j,t-1}}$$

To avoid leverage from outliers driven by either anomalous growth or administrative errors in data entry, growth values are top-coded at a ratio of 2 (i.e., extraction is assumed to not grow at a rate larger than 100% per annum):

$$\hat{Q}^{(r)}_{j,t} = \min\left(\hat{Q}^{(r)}_{j,t}, 2\right)$$

Between 0 and 265 values for gross growth rates are winsorized for each resource. Observations affected by missing data or zero base-period values are normalized to one. After these adjustments, the dataset has 11,067 entries, a balanced panel of 217 geographic entities between 1970 and 2020.

A majority (between 6,480 and 9,942) of entries are initially missing for all resources and replaced with a gross growth rate of one.

3.4 Törnqvist Weights

Non-renewable resources: Total rents paid to non-renewable natural capital each period in country i can be calculated as

$$R_{j,t}^{NR} = \sum_{r \in \mathcal{N}} R_{j,t}^{(r)}$$

where $\mathcal{N}$ is the set of non-renewable resources given in Table 11. The value share for each resource (its share of total rents across all non-renewable resources that period) is calculated using:

$$w_{j,t}^{(r)} = \frac{R_{j,t}^{(r)}}{R_{j,t}^{NR}}$$

These value shares are used to construct period-t Törnqvist weights based on an equally-weighted average of rent shares in period t and $t-1$

$$\tilde{w}_{j,t}^{(r)} = \frac{1}{2}\left(w_{j,t}^{(r)} + w_{j,t-1}^{(r)}\right)$$

For non-renewable resources, we are able to construct rent weights for 6,064 observations at the country level, an unbalanced panel across all 217 countries.

Renewable resources: Total renewable rent each year is defined analogously as

$$R_{j,t}^{R} = \sum_{k \in \mathcal{R}} R_{j,t}^{(k)}$$

and weights in each year are are computed using:

$$w_{j,t}^{(k)} = \frac{R_{j,t}^{(k)}}{R_{j,t}^{R}}$$

For the construction of this index, urban land is assigned a wealth value equal to that of agricultural land as opposed to one that is 1/1.24 of the produced capital stock following Kunte et al. For renewable resources, we are able to construct rent weights for 3,684 observations at the country level, an unbalanced panel across 149 countries between 1995 and 2020. Because this leaves us with a substantial set of observations where quantities are observed but rent weights are missing, each resource is instead assigned the average value of weights across all years where data are available

$$\bar{w}_{j}^{(k)} = \frac{1}{T}\sum_{t=1}^{T} w_{j,t}^{(k)}$$

for all time when constructing the indices for renewable inputs.

3.5 Törnqvist Indices

Non-renewable resources: For each resource r in the set of non-renewables $\mathcal{N}$, individual gross changes in quantities are then exponentiated:

$$Q_{j,t}^{(r)*} = \left(\hat{Q}_{j,t}^{(r)}\right)^{\tilde{w}_{j,t}^{(r)}}$$

The composite Törnqvist quantity index for non-renewable inputs $\mathcal{Q}_{j,t}^{N}$ is the product across resources of all exponentiated terms.

$$\mathcal{Q}_{j,t}^{N} = \prod_{r\in\mathcal{N}} Q_{j,t}^{(r)*}$$

Growth in the non-renewable index is then defined as

$$g_{j,t}^{NR} \triangleq \mathcal{Q}_{j,t}^{N} - 1$$

when used for calculating in-sample means, standard deviations, and correlations for non-renewable resource use growth at the national level.

Renewable resources: The renewable Törnqvist index $\mathcal{Q}_{j,t}^{R}$ is constructed in an analogous manner. First, gross quantity changes for all renewable resources k are exponentiated using the time-invariant rent share weights

$$Q_{j,t}^{(k)*} = \left(Q_{j,t}^{(k)}\right)^{\bar{w}_{i}^{(k)}}$$

and the index is computed as

$$\mathcal{Q}_{j,t}^{R} = \prod_{k\in\mathcal{R}} Q_{j,t}^{(k)*}$$

Growth in the renewable natural capital Törnqvist quantity index is then set as

$$g_{j,t}^{R} \triangleq \mathcal{Q}_{j,t}^{R} - 1$$

when used for calculating correlations at the national level.

Finally, both $g_{j,t}^{R}$ and $g_{j,t}^{NR}$ are topcoded at 1, ruling out over 100% growth in the quantity indices. This affects 86 instances of the renewables index and zero instances of the non-renewables index.

3.6 Regression Analysis

The Törnqvist indices for renewable and non-renewable capital inputs are intended to capture factor input growth the national level. We know from common knowledge that these inputs, especially non-renewable hydrocarbons, should explain some degree of output growth. To ensure growth in our indices for the use of non-renewable and renewable capital are positively associated with output, we run the following regression

$$\Delta \ln(y_{j,t}) = \gamma_1 \, g_{j,t}^{NR} + \gamma_2 \, g_{j,t}^{R} + \sum_{x \in \boldsymbol{Z}} \alpha_z \, g_{z_{j,t}} + \delta_j + \zeta_t + \varepsilon_{j,t} \tag{18}$$

where $y_{j,t}$ is the logarithm of real GDP at constant national prices taken from the PWT intended for use measuring GDP growth (Feenstra, Inklaar, and Timmer). The set $\boldsymbol{Z} = \{K, L, HC, LS\}$ contains controls for growth in produced capital, employment, human capital, and the labor share in each country each year. Terms (δ_j, ζ_t) are again two-way fixed effects, and $\varepsilon_{j,t}$ are idiosyncratic error terms. The coefficients γ_1 and γ_2—output elasticities for our indices of non-renewable and renewable capital—are our estimands of interest and the feasible analogs of the theoretical counterparts in the simulation section.

Table 13: TFP Growth vs. Natural Capital Growth

$\Delta \ln(y_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
$g_{j,t}^{R}$	0.130	0.073	0.100	0.064	0.066	0.047
	(0.049)	(0.040)	(0.039)	(0.037)	(0.037)	(0.016)
$g_{j,t}^{NR}$	0.025	0.013	0.017	0.011	0.011	0.013
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
# Observations	4974	3953	4974	3953	3953	3697
# Countries	134	104	134	104	104	104

Note: Standard errors clustered at the country level are shown in parentheses. Rows under the "Controls for ... " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable and is estimated using the Arellano-Bond (1991) dynamic panel estimator.

Table 13 shows the resulting estimates of γ_1, γ_2 from five variants of the specification in equation 18 when estimated on an unbalanced panel of either 104 or 134 countries between 1972 and 2019 (the number of observations and covered countries depends on the availability of controls). Consistent with our priors that growth in the use of natural capital would be associated with output growth, estimates of output elasticities γ are positive across all specifications for growth in both input indices. Point estimates for the output elasticities of non-renewable and renewable natural capital services hover around 0.02 and 0.06 respectively. For non-renewables, the estimates are significant at traditional levels across all specifications ($p < 0.01$) and comparable in magnitude to the output elasticity of energy (which is typically composed mainly of fossil fuels) found in the literature using Cobb-Douglas specifications for aggregate output (see Golosov et al.; Casey). The coefficients on renewables, while larger in magnitude, are less precise with p-values ranging from 0.01 to 0.13 depending on specification. Both variables also survive a standard 10-fold cross-validation procedure for specification selection. Given the results in Table 13 for $\hat{\gamma}_2$, we assign γ_2 as 0.05 when calibrating the simulations described in Section 4.

4 Country-Level Simulation Exercise

We now describe the simulation exercise underpinning the results in Section 3 of the main text.

4.1 Setup

Data Generating Process for Output Starting with the model of Section 1.4, we again assume output growth in each country is comprised of growth in productivity, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vector $\boldsymbol{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j. Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \boldsymbol{g}'_{j,t}\boldsymbol{\gamma}_j \tag{19}$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for that country.

Structural Parameters We set $\gamma_2 = 0.05$ based on the analysis in Section 3.6. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al.), we set γ_1 as 0.05. This ad-hoc choice is slightly lower than our point estimates in Section 3.6, but in line with estimates of the factor share for fossil energy in the United States since 1947 (Hassler, Krusell, and Olovsson).[10]

Procedure for Estimating TFP Growth We assume an econometrician observes a subset of growth in factor inputs—those for K, L, and N_1—along with growth in output without error for 25 periods (years) in each country. We select a sample size of 25 periods because the CWON data on natural capital cover this many years in each country. She deploys the model estimating TFP growth while omitting non-renewable resources (model 1) by estimating:

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \tag{20}$$

on her time series of data for growth in $\{K, L\}$ in a given country using ordinary least squares (OLS). Her estimator for average TFP growth under model 1 is the intercept term $\hat{\gamma}_A$. She estimates the analog for the case including non-renewable resources (model 2)

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \tag{21}$$

on time series for growth in $\{K, L, N_1\}$. The OLS-based estimator for average TFP growth under model 2 is $\tilde{\gamma}_A$.

[10] Note we need not make assumptions on γ_L and γ_K in order to solve for the bias or mean squared error of the intercept term.

Parameters Governing Growth in Inputs We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. This restriction can be relaxed to allow arbitrary correlations between growth in factor inputs and true TFP growth, but we elect to set these correlations as zero in all countries as we do not have strong priors for what these unobservable correlations should be. The simulation procedure would remain virtually unchanged in this case apart from assigning alternative values to those off-diagonal terms in Σ_j for each country.

Calculating the bias and mean squared error for the TFP estimators from on models 1 and 2 requires knowing the mean ($\boldsymbol{\mu}_j$) and variance (Σ_j) governing the joint distribution of growth terms $\boldsymbol{g}_j$ in each country.[11] The moments and the sources we use to calibrate them are listed in the first column of Table 14. Columns 2 and 3 of Table 14 list respectively the sample analogs and the source data used to construct them. We set the parameters governing average growth in the use of natural capital equal to the sample means for the growth in renewable and non-renewable Törnqvist indices constructed in Section 3 between 1996 and 2019 (Table 14 rows one and two). Rows three, four, and five—governing expected growth in produced capital services, labor, and TFP—are set equal to their sample analogs in the Penn World Tables (Feenstra, Inklaar, and Timmer) over the same 1996-2019 period. Finally, higher order moments are set based on the empirical covariances between growth in our Törnqvist non-renewable index and growth in other inputs we take from the PWT during the 1996-2019 period.

Table 14: Simulation Moments and Sources

Population Moment	**Sample Moment**	**Source**
$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON (Sec. 3)
$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON (Sec. 3)
$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Σ	$\widehat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the VCE matrix Σ. CWON refers to the World Bank *Changing Wealth of Nations* database (World Bank), and PWT refers to the Penn World Table (Feenstra, Inklaar, and Timmer).

Given the limited overlap in data between the CWON and PWT data, we are ultimately able calculate sample analogs of $\boldsymbol{\mu}$ and Σ for 100 countries which we substitute directly into the formulas for bias and RMSE below.

[11] Technically, we need not know $\mathbb{E}[g_{TFP}]$ to calculate the bias or expected MSE. However, we do use this to calculate a ballpark ratio of bias reduction to true underlying growth so it's treated as needed here.

4.2 Bias in Estimated TFP

We now turn to calculating the bias and mean squared error for each estimator at the country level. As shown in Section 6.4, when g_{TFP} is independent and identically distributed, the bias for the estimate of average TFP growth from model 1 is

$$\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]] = \gamma_1 \mathbb{E}[g_{N_1} - \kappa_{N_1,\tilde{K}} g_K - \kappa_{N_1,\tilde{K}} g_L] + \gamma_2 \mathbb{E}[g_{N_2} - \kappa_{N_2,\tilde{K}} g_K - \kappa_{N_2,\tilde{K}} g_L] \tag{22}$$

where the $\kappa_{N_j,\cdot}$ coefficients are generated by the projections of g_{N_1} and g_{N_2} onto the vector of included factor growth $\hat{\boldsymbol{g}} = [g_K, g_L]'$:

$$\boldsymbol{\kappa}_{N_1} = \mathbb{E}[\hat{\boldsymbol{g}}\, \hat{\boldsymbol{g}}']^{-1} \mathbb{E}[\hat{\boldsymbol{g}}\, g_{N_1}]$$
$$\boldsymbol{\kappa}_{N_2} = \mathbb{E}[\hat{\boldsymbol{g}}\, \hat{\boldsymbol{g}}']^{-1} \mathbb{E}[\hat{\boldsymbol{g}}\, g_{N_2}]$$

For the case of Model 2 where only N_2 is omitted, the bias is

$$\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]] = \gamma_2 \mathbb{E}[g_{N_2} - \lambda_{N_2,\tilde{K}} g_K - \lambda_{N_2,\tilde{L}} g_L - \lambda_{N_2,\tilde{N}_1} g_{N_1}] \tag{23}$$

where the $\lambda_{N_2,\cdot}$ coefficients are generated by the projection of g_{N_2} onto $\tilde{\boldsymbol{g}} = [g_K, g_L, g_{N_1}]'$. :

$$\boldsymbol{\lambda} = \mathbb{E}[\tilde{\boldsymbol{g}}\, \tilde{\boldsymbol{g}}']^{-1} \mathbb{E}[\tilde{\boldsymbol{g}}\, g_{N_2}]$$

4.3 Expected Improvements in Root Mean Squared Error

Under the assumption of homoskedastic errors (constant variance in underlying TFP growth across observations) and independent sampling of growth terms, Section 6.5 shows the estimator $\hat{\gamma}$ in (20) has conditional variance

$$\mathrm{Var}(\hat{\gamma}|\hat{\boldsymbol{g}}) = \left(\sigma^2_{TFP} + \begin{bmatrix}\gamma_1, \gamma_2\end{bmatrix} \Sigma_{1,2|\boldsymbol{g}} \begin{bmatrix}\gamma_1 \\ \gamma_2\end{bmatrix}\right) (\hat{\boldsymbol{g}}'\hat{\boldsymbol{g}})^{-1}$$

where here in a slight abuse of notation, $\hat{\boldsymbol{g}} = [1, g_K, g_L]$ is the 25×3 matrix of observed growth terms (and a constant) included under model 1 and $\Sigma_{1,2}$ is the (conditional) covariance matrix for g_{N_1} and g_{N_2}. The conditional variance of of the intercept term is then

$$\mathrm{Var}(\hat{\gamma}_A|\hat{\boldsymbol{g}}) = \frac{\sigma^2_{TFP} + \gamma_1^2\sigma_1^2 + \gamma_2^2\sigma_2^2 + 2\gamma_1\gamma_2\sigma_{1,2}}{25}$$

where $\sigma^2_{\cdot}$ and $\sigma_{1,2}$ are the (conditional) variances and covariance of growth in the natural capital stocks.

Section 6.5 shows the estimator of TFP from model 2 has variance

$$\text{Var}(\tilde{\gamma}_A|\tilde{\boldsymbol{g}}) = \frac{\sigma^2_{TFP} + \gamma_2^2\sigma_2^2}{25}$$

where $\tilde{\boldsymbol{g}} = [1, g_K, g_L, g_{N_1}]$ is the matrix of regressors under Model 2.

We calculate the expected change in root mean squared error term—Δ RMSE—as

$$\Delta\text{RMSE} \triangleq \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\hat{\boldsymbol{g}}]} - \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\tilde{\boldsymbol{g}}]} \tag{24}$$

which can be decomposed as

$$\Delta\text{RMSE} \triangleq \sqrt{\text{Var}(\hat{\gamma}_A|\hat{\boldsymbol{g}}) + \mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]|\hat{\boldsymbol{g}}]^2} - \sqrt{\text{Var}(\tilde{\gamma}_A|\tilde{\boldsymbol{g}}) + \mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]|\tilde{\boldsymbol{g}}]^2} \tag{25}$$

as $\mathbb{E}[g_{TFP}]$ is a constant and growth in TFP is assumed iid. The (squared) bias terms δ and ε are defined in Section 4.2 and become biases conditional on the observed portion of $\boldsymbol{g}$ when placed inside the operator $\mathbb{E}[\cdot|\boldsymbol{g}]$.

4.4 Results

4.4.1 Decomposition of RMSE Improvement

As referenced in Section 3 of the main text, the vast majority of RMSE in estimated TFP growth under both models 1 and 2 stems from the bias terms. Figure 2 gives a histogram of the share of RMSE attributable to the bias terms

$$\text{Share 1} = \frac{\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]|\hat{\boldsymbol{g}}]^2}{\text{Var}(\hat{\gamma}_A|\hat{\boldsymbol{g}}) + \mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]|\hat{\boldsymbol{g}}]^2}$$

for the model omitting both NK stocks and

$$\text{Share 2} = \frac{\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]|\tilde{\boldsymbol{g}}]^2}{\text{Var}(\tilde{\gamma}_A|\tilde{\boldsymbol{g}}) + \mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]|\tilde{\boldsymbol{g}}]^2}$$

for the model including nonrenewables.

Figure 3 gives a sense of the reduction in (squared) bias and the corresponding (lack of) reduction in variance when going from the estimator $\hat{\gamma}_A$ to $\tilde{\gamma}_A$.

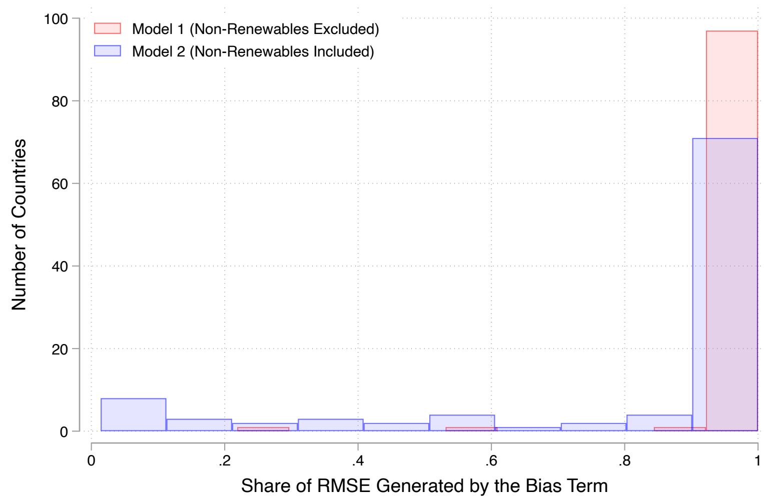

Figure 2: Histograms showing the share of the RMSE terms under each model attributed to the (squared) bias terms $\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]]^2$ and $\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]]^2$.

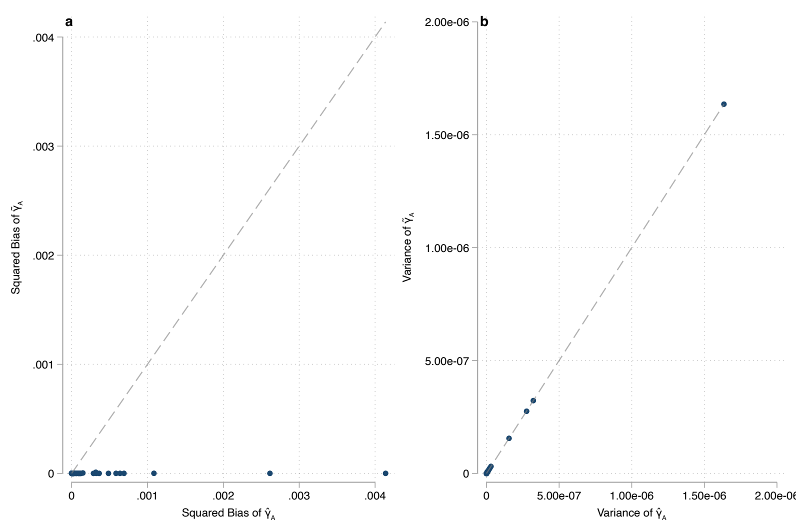

Figure 3: Panel 3a shows country-level biases in $\tilde{\gamma}_A$ and $\tilde{\gamma}_A$. Panel 3b shows their variances. Dashed lines are drawn along the 45 degree line where the squared biases or variances are

4.4.2 Constructing the Country-Level Overlay on Figure 1

Table 15 shows data for the set of 100 individual countries overlain on Figure 1 (plotted in Figure 2) in the main text. The two columns labeled bias and RMSE Reduction denote the effect of adopting an estimator including non-renewable resources on the accuracy of the TFP estimate at the country level in unconditional expectation and conditional on observed data.

Table 15: Country-Level Sample Moments and Simulation Results

ISO3 Code	Country	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
AGO	Angola	0.04	0.01	0.13	0.07	9.25e-04	1.02e-04	0.02
ARG	Argentina	-0.01	-0.00	0.13	-0.15	8.44e-04	5.29e-05	-0.00
ARM	Armenia	0.16	0.01	0.16	0.19	9.03e-03	2.90e-03	0.06
AUS	Australia	0.05	-0.01	-0.13	0.08	2.75e-03	1.55e-03	0.00
AUT	Austria	-0.02	0.00	-0.08	0.09	-1.78e-04	3.05e-06	0.00
BDI	Burundi	0.08	-0.00	0.04	0.03	4.94e-03	2.09e-03	-0.01
BEL	Belgium	0.22	-0.00	0.12	-0.56	3.99e-03	1.05e-03	0.00
BEN	Benin	-0.04	0.01	-0.16	-0.04	2.76e-03	6.61e-04	0.02
BFA	Burkina Faso	0.21	0.00	0.06	-0.10	2.09e-03	4.25e-04	0.01
BGR	Bulgaria	-0.01	-0.00	-0.27	0.14	1.26e-03	9.79e-05	-0.01
BHR	Bahrain	0.04	-0.00	0.13	-0.00	2.07e-03	7.52e-04	-0.01
BOL	Bolivia	0.04	-0.00	-0.19	-0.12	1.01e-03	1.86e-04	0.00
BRA	Brazil	0.06	0.01	0.15	0.08	1.29e-03	2.94e-04	-0.01
BWA	Botswana	0.01	-0.00	-0.08	-0.16	4.01e-03	9.23e-04	-0.01
CAF	Central African Republic	0.08	0.00	0.31	-0.15	3.95e-03	4.07e-04	-0.00
CAN	Canada	0.02	0.00	0.13	-0.04	4.30e-04	6.54e-05	0.00
CHE	Switzerland	0.15	0.00	-0.03	-0.51	2.63e-03	1.55e-03	0.01
CHL	Chile	0.03	0.00	0.30	-0.42	-1.42e-03	-3.35e-04	-0.00
CHN	China	0.04	0.00	0.54	-0.01	7.86e-03	4.37e-03	0.01
CIV	Côte d'Ivoire	0.09	0.00	-0.12	-0.28	2.04e-03	3.20e-04	0.01
CMR	Cameroon	0.02	0.00	-0.24	0.45	1.69e-02	1.33e-02	0.01
COL	Colombia	0.02	0.00	0.40	0.04	9.08e-04	1.77e-04	-0.00
CRI	Costa Rica	0.18	-0.00	-0.15	-0.03	2.18e-02	1.62e-02	0.01
CZE	Czech Republic	-0.01	-0.00	0.22	-0.43	1.26e-03	2.23e-04	0.01
DEU	Germany	-0.03	-0.00	0.37	-0.14	3.82e-03	2.10e-03	0.00
DNK	Denmark	-0.03	-0.00	0.31	-0.55	1.09e-02	8.08e-03	0.00
DOM	Dominican Republic	-0.01	-0.00	0.04	-0.21	3.61e-03	1.05e-03	0.02
ECU	Ecuador	0.01	0.01	0.01	-0.20	-3.96e-04	-4.10e-05	0.00
EGY	Egypt	-0.00	0.00	0.09	0.38	1.36e-04	2.46e-05	-0.01
ESP	Spain	-0.02	-0.00	-0.49	0.05	8.95e-03	7.23e-03	-0.00
EST	Estonia	0.06	0.00	0.24	0.68	2.90e-03	6.21e-04	0.02
FIN	Finland	0.07	-0.00	0.19	0.28	1.38e-03	3.56e-04	0.01
FRA	France	-0.07	-0.00	0.04	0.19	4.21e-03	2.60e-03	0.00
GAB	Gabon	-0.03	-0.00	0.10	-0.36	4.76e-04	1.76e-05	-0.01
GBR	United Kingdom	-0.05	0.00	0.46	-0.20	7.51e-03	5.76e-03	0.00
GRC	Greece	-0.02	-0.01	0.20	-0.06	2.06e-03	6.20e-04	-0.00
GTM	Guatemala	0.02	-0.00	-0.03	0.08	9.05e-03	5.60e-03	0.00
HND	Honduras	0.02	-0.00	-0.13	0.02	2.07e-03	3.71e-04	-0.00

ISO3 Code	Country	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
HRV	Croatia	-0.01	-0.01	0.07	-0.04	2.85e-03	1.17e-03	0.01
HUN	Hungary	-0.04	-0.00	-0.06	0.11	1.69e-03	4.54e-04	0.01
IDN	Indonesia	0.01	0.01	0.16	-0.06	-9.80e-05	-7.29e-06	0.00
IND	India	0.03	-0.00	-0.03	-0.01	9.94e-04	1.35e-04	0.02
IRL	Ireland	-0.03	0.00	-0.03	0.10	6.23e-04	2.49e-05	0.01
IRN	Iran	0.00	-0.01	0.20	-0.08	1.95e-03	2.33e-04	-0.01
IRQ	Iraq	0.10	0.00	-0.34	-0.23	3.63e-03	2.31e-04	0.03
ISR	Israel	0.10	0.00	-0.59	-0.31	2.60e-02	2.23e-02	0.00
ITA	Italy	-0.04	-0.00	-0.07	-0.05	1.51e-03	5.50e-04	-0.01
JAM	Jamaica	-0.00	-0.00	0.35	-0.18	2.08e-03	5.72e-04	-0.01
JOR	Jordan	0.03	-0.00	-0.04	-0.10	1.46e-03	2.98e-04	-0.00
JPN	Japan	-0.02	-0.00	-0.10	-0.24	4.12e-04	5.22e-05	0.00
KAZ	Kazakhstan	0.02	0.00	-0.05	-0.22	2.52e-03	4.24e-04	0.03
KEN	Kenya	0.15	-0.00	-0.10	-0.00	5.08e-02	4.41e-02	-0.00
KGZ	Kyrgyzstan	0.08	-0.00	-0.38	0.22	1.90e-02	1.31e-02	0.02
KOR	South Korea	-0.08	-0.01	0.06	0.19	7.48e-03	4.44e-03	0.01
KWT	Kuwait	0.03	0.00	0.07	0.21	1.32e-03	8.84e-05	-0.03
LAO	Laos	0.19	0.01	-0.04	0.33	3.22e-02	2.72e-02	0.00
LKA	Sri Lanka	0.01	0.00	0.14	0.20	3.91e-03	1.91e-03	0.00
LTU	Lithuania	0.00	-0.00	-0.19	-0.46	3.32e-03	8.30e-04	0.02
MAR	Morocco	0.01	-0.00	-0.29	-0.14	9.81e-03	5.69e-03	0.01
MEX	Mexico	-0.01	-0.00	0.16	0.36	2.05e-03	6.52e-04	-0.00
MNG	Mongolia	0.09	-0.00	-0.06	0.15	3.61e-03	1.03e-03	0.02
MOZ	Mozambique	0.14	0.02	0.06	-0.09	1.47e-02	1.00e-02	0.01
MRT	Mauritania	-0.00	-0.00	0.14	0.45	2.74e-03	4.20e-04	-0.00
MYS	Malaysia	0.01	0.00	0.50	0.08	7.77e-04	7.06e-05	0.01
NAM	Namibia	0.02	-0.00	-0.05	-0.03	1.11e-03	1.38e-04	-0.00
NER	Niger	0.01	0.01	0.06	0.17	2.49e-02	1.84e-02	0.01
NGA	Nigeria	0.01	0.00	0.40	0.01	1.84e-03	2.48e-04	0.02
NIC	Nicaragua	0.11	-0.00	-0.18	-0.09	1.06e-02	7.25e-03	-0.00
NLD	Netherlands	-0.03	-0.00	0.02	0.05	3.29e-03	1.80e-03	0.00
NOR	Norway	0.00	-0.00	-0.02	-0.06	5.64e-03	3.59e-03	-0.00
NZL	New Zealand	0.01	-0.01	0.08	0.07	8.75e-04	2.81e-04	0.00
PAN	Panama	0.12	0.01	-0.00	-0.33	1.73e-03	1.94e-04	-0.00
PER	Peru	0.05	0.01	-0.39	-0.12	7.38e-03	4.95e-03	-0.00
PHL	Philippines	0.07	0.00	-0.26	-0.09	9.68e-03	6.82e-03	0.01
POL	Poland	0.00	-0.00	0.11	0.18	9.37e-04	2.55e-04	0.02
PRT	Portugal	-0.00	-0.00	-0.28	-0.04	2.10e-03	8.97e-04	-0.00
QAT	Qatar	0.06	0.00	0.32	0.02	-3.67e-04	-9.50e-06	-0.03
ROU	Romania	-0.06	-0.00	0.13	-0.22	3.80e-03	9.17e-04	0.02
RUS	Russia	-0.03	-0.00	-0.02	-0.08	1.16e-03	1.02e-04	0.02
RWA	Rwanda	0.18	0.01	0.16	-0.51	1.03e-02	5.34e-03	0.03
SAU	Saudi Arabia	0.07	0.00	0.07	-0.29	2.40e-03	7.21e-04	-0.02
SDN	Sudan	0.14	-0.01	0.25	0.52	4.24e-03	1.45e-03	0.01
SEN	Senegal	0.07	-0.00	0.32	0.10	5.50e-03	2.51e-03	0.00
SLE	Sierra Leone	0.14	0.00	0.17	0.03	1.02e-02	2.25e-03	-0.00
SRB	Serbia	-0.03	-0.00	0.25	-0.28	3.59e-03	6.91e-04	0.03

ISO3 Code	Country	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
SVK	Slovakia	-0.04	-0.00	0.03	-0.06	2.25e-03	6.75e-04	0.02
SVN	Slovenia	-0.03	0.01	-0.05	-0.35	9.21e-04	1.03e-04	0.01
SWE	Sweden	0.03	-0.00	0.11	0.20	-3.12e-05	1.74e-05	0.01
SWZ	Eswatini	0.03	0.00	0.15	-0.11	5.01e-03	1.86e-03	0.01
TGO	Togo	0.09	0.00	-0.25	0.18	1.78e-02	1.09e-02	0.02
THA	Thailand	0.05	0.00	0.32	-0.24	7.36e-04	4.03e-05	0.02
TTO	Trinidad and Tobago	0.01	-0.01	-0.05	0.17	4.76e-04	1.21e-05	0.00
TUN	Tunisia	-0.01	0.00	0.34	-0.06	4.78e-03	2.94e-03	0.01
TUR	Turkey	0.04	0.00	0.06	0.22	1.31e-03	1.82e-04	-0.00
TZA	Tanzania	0.12	0.00	-0.43	0.07	2.40e-02	2.02e-02	0.01
URY	Uruguay	-0.07	0.02	0.08	-0.14	4.51e-03	2.60e-03	0.01
USA	United States	0.02	-0.00	-0.39	-0.23	3.69e-03	2.58e-03	0.01
ZAF	South Africa	0.00	-0.00	0.24	0.31	1.33e-03	2.50e-04	-0.00
ZMB	Zambia	0.04	0.00	0.08	0.23	1.64e-03	2.66e-04	0.02
ZWE	Zimbabwe	0.00	0.01	0.32	0.40	-6.74e-04	8.91e-07	0.01

4.4.3 Cross Sectional Analysis of Simulation Results

This section shows additional cross-sectional analysis of the RMSE improvements from the simulation exercise in Section 3 of the main text. Figure 4 plots RMSE reductions against four additional statistics at the country level measured in 2019: (log) GDP, (log) GDP per capita, human capital levels, and statistical capacity. Panels 4a and 4b indicate that the magnitude of RMSE reductions tends to be decreasing in both measures of income. The relationship is robust and fairly substantial in magnitude; a three-fold increase in GDP decreases estimation error in TFP growth by 20 basis points (double our baseline estimate). A similar relationship holds between these reductions and the level of human capital in a given country (Panel 4c), which is unsurprising given the correlation between education and output. This relationship can be explained by higher-income countries having smaller and less-volatile TFP growth, which lowers the variance of both estimators and thus shrinks the error reduction in absolute terms.

Panel 4d shows the relationship between RMSE improvement and statistical capacity as measured by the World Bank for 67 developing countries in our sample. The correlation here is a fairly precise zero, which we interpret as suggestive evidence that better measurement of natural capital may benefit countries regardless of current national accounting practices. The last two panels take an alternative direction and examines how including non-renewables in our simulations relates to two measures of the *level* of resource intensity in production at the country-level. Panel 4e shows RMSE reductions against resource rents as share of GDP taken from the World Bank's World Development Indicators database (World Bank). Panel 4f plots the improvements against an alternative metric for natural capital intensity, resource depletion as a share of gross national income. Perhaps surprisingly, there is a precise negative (albeit small in magnitude) statistical relationship between both measures of resource dependence and error reduction. For highly resource-dependent economies (e.g., Iraq and Kuwait), RMSE reductions are some of the smallest in our sample (1 and 3 basis points respectively). This negative relationship persists qualitatively after controlling for the other factors that have negative correlations shown here (i.e., human capital and GDP), but loses statistical significance and flips signs when conditioned on historical volatility in TFP (the driver of the variance portion of RMSE spoken to above).

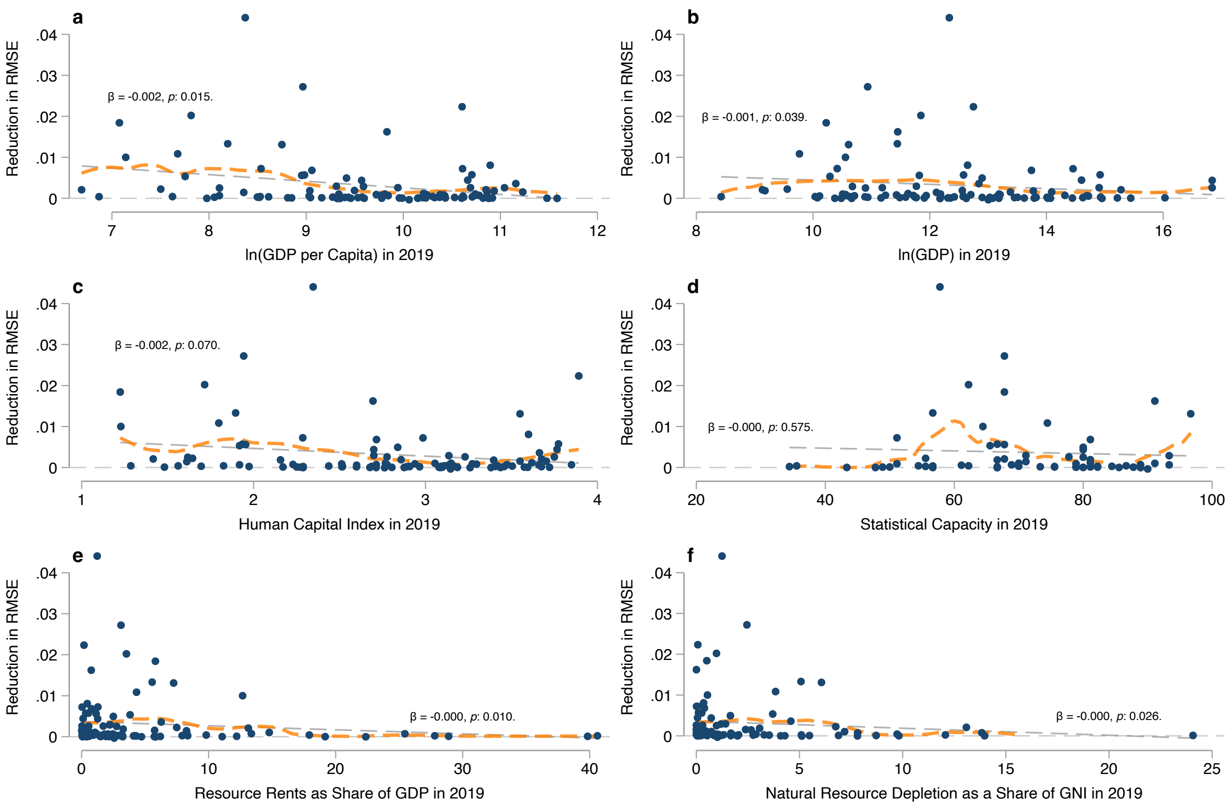

Figure 4: Panels 4a and 4b plot reductions in the RMSE of estimated TFP growth against the (logarithm) of GDP per capita and aggregate GDP at the country level. Panels 4c and 4d plots these improvements against measures of human capital and statistical capacity take from the Penn World Tables (Feenstra, Inklaar, and Timmer). Panels 4e and 4f show RMSE reductions against the share of GDP that comes from resource rents and the share of national income attributable to natural resource depletion, respectively. Dashed lines show fits from a simple linear regressions and the overlain text shows the $\hat{\beta}$ coefficient from the linear regression along with the associated p-values. Dashed orange lines shows fitted values from a local linear regression (Fan).

4.4.4 Reduced-Form Attribution of RMSE Improvement

Table 16 shows the regression coefficients when ΔRMSE is projected onto the elements of the sample versions of $\boldsymbol{\mu}_j$ and Σ from Section 4.3. The test statistic for the joint null hypothesis of the coefficients on $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$ in Table 16 both being zero is 0.039. When the regression is restricted to a bivariate one containing only $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$, the test statistic for the joint null falls below 0.01. $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$ are also retained as regressor when applying a 10-fold cross validation LASSO procedure. Figure 5 gives a visual version of each regression coefficient by plotting residualized values ΔRMSE against the 12 different sample moments in each country. The slopes of the dashed black lines in Panels a-l correspond to the regression coefficients in Table 16.

Table 16: RMSE Improvement Decomposition

Dep. Var: ΔRMSE	(1)
$\rho(g_{N_1}, g_K)$	-0.006
	(0.003)
$\rho(g_{N_1}, g_{N_2})$	0.003
	(0.002)
$\rho(g_{N_2}, g_L)$	-0.001
	(0.003)
$\rho(g_{N_1}, g_L)$	-0.007
	(0.003)
$\rho(g_L, g_K)$	-0.004
	(0.002)
$\rho(g_{N_2}, g_K)$	0.001
	(0.003)
$\bar{g}_L$	-0.014
	(0.057)
$\bar{g}_{N_1}$	0.039
	(0.010)
$\bar{g}_K$	0.007
	(0.033)
$\bar{g}_{N_2}$	0.204
	(0.137)
$\bar{g}_A$	-0.034
	(0.059)
σ_A	-0.057
	(0.025)
R^2	0.38
N	100

Standard errors shown in parentheses.

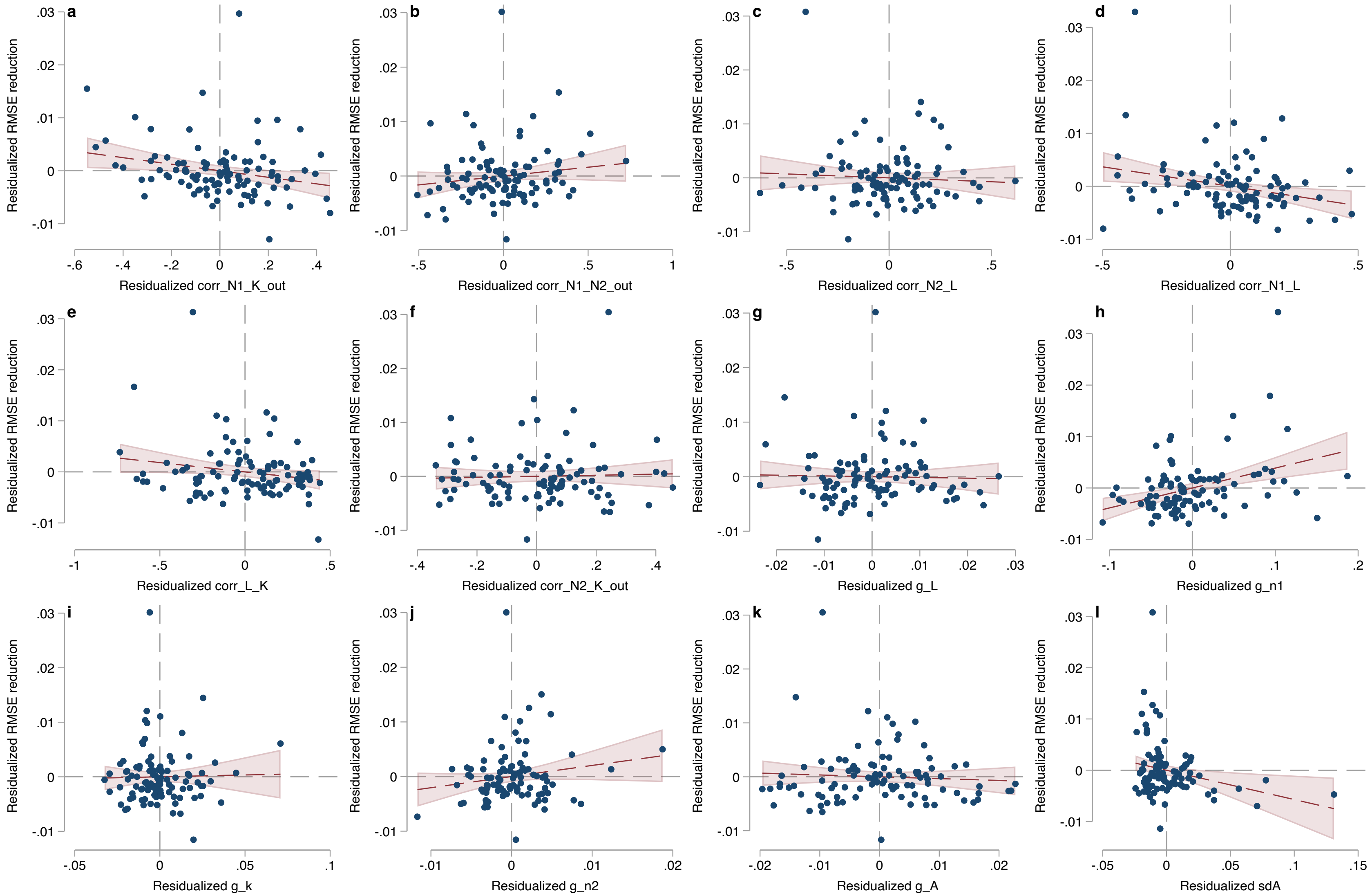

Figure 5: Frisch-Waugh-Lovell plot of residualized ΔRMSE against residualized versions of each of the 12 sample moments that determine its value at the country level. Dashed black lines show the linear regression coefficients in Table 16 while the red shaded regions denote 9 percent confidence intervals of the fitted values.

4.5 Sensitivity Analysis of Simulation Results

The applicability of our simulation results in Section 3 of the main text rely on our choices of parameters governing growth in output and factor inputs accurately reflecting reality. While our choice of parameters is based on sample moments (i.e., Table 14), our own econometric analysis (i.e., Table 13), or estimates from prior literature (e.g., Arrow et al.; Hassler, Krusell, and Olovsson), they play a critical role in determining whether including non-renewables improves estimates of TFP. In this section we test the sensitivity of our simulation results on two fronts: choices of output elasticities for natural capital (γ_1 and γ_2) and the sample periods for estimating the growth rates of inputs at the country level ($\boldsymbol{\mu}_j$ and Σ_j).

4.5.1 Choice of Output Elasticities

Section 4.5.1 tests the sensitivity of our results to choices of elasticities γ_1 and γ_2. To do this, we run a Monte Carlo analysis over draws of random pairs for alternative outptu elasticities drawn from iid uniform distributions ($\mathcal{U}[0, 0.1] \times \mathcal{U}[0, 0.1]$) where each pair is set as an alternative output elasticity for non-renewables (renewables) for a random country in our sample. We store the resulting ΔRMSE for country j under these alternative assumptions for γ_1, γ_2 holding all other relevant moments in ($\boldsymbol{\mu}_j$ and Σ_j) unchanged. We repeat this process 10,000,000 times (roughly 100,000 times per country).

Figure 6 shows a histogram of the results from the Monte Carlo exercise. Each individual MC draw produces an alternative ΔRMSE_j for a given country given a random pair of output elasticities for natural capital, γ_1, γ_2, drawn from $\mathcal{U}[0, 0.1]$. The median (mean) reduction in RMSE at the country level across our 10,000,000 alternative draws falls to 2 (28) basis points. The distribution remains quite disperse, displaying a standard deviation of 80 basis points. Unlike with our main analysis, a substantial share (31 percent) of country-level outcomes are negative under random alternative elasticities (as shown by the substantial mass of the histogram to the left of the vertical dashed line at zero).

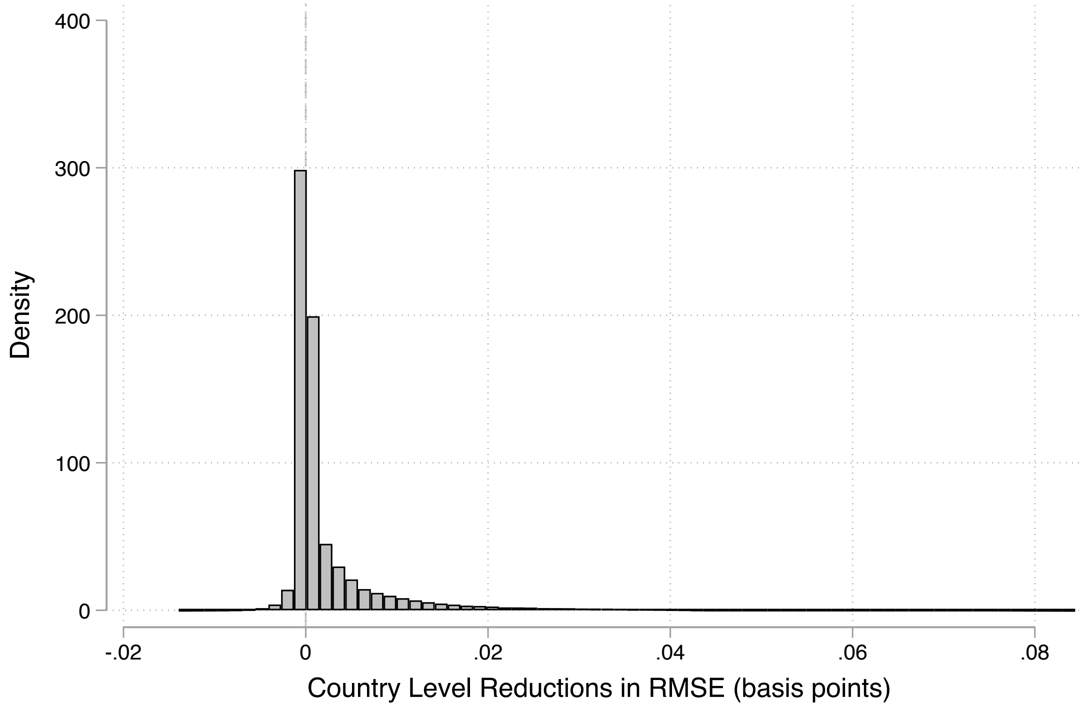

Figure 6: Histogram of ΔRMSE_j across each of $N = 10,000,000$ Monte Carlo draws of alternative output elasticity pairs γ_1, γ_2 at the country level.

4.5.2 Choice of Sample Period

To test the sensitivity of our results to the choice of sample period for constructing the sample analogs of population moments $\boldsymbol{\mu}_j$ and Σ_j, we split our 1996-2019 sample into early (1996-2008) and late (2009-2019) subsample periods and repeat the simulation exercise. This reduces the set of countries from 100 to 93 and 98 respectively in each sample period due to data limitations.

For the early period (1996-2008), the median (mean) reduction in RMSE rises to 16 (84) basis points. The results are also substantially more disperse as the sample versions of Σ become fairly large for the small number of periods; the standard deviation of ΔRMSE rises to 155 and 200 basis points for the eraly and late periods, respectively. Ten countries display increases in RMSE when going from model 1 to model 2 in the early period, while four display increases in the late period. changes in Figures 7 and 8 reconstruct alternative versions of Figure 2 in the main text for each subsample. The qualitative nature of our results—that including natural capital on the margin substantially improves estimates of TFP growth for most countries—remains true for both subsamples.

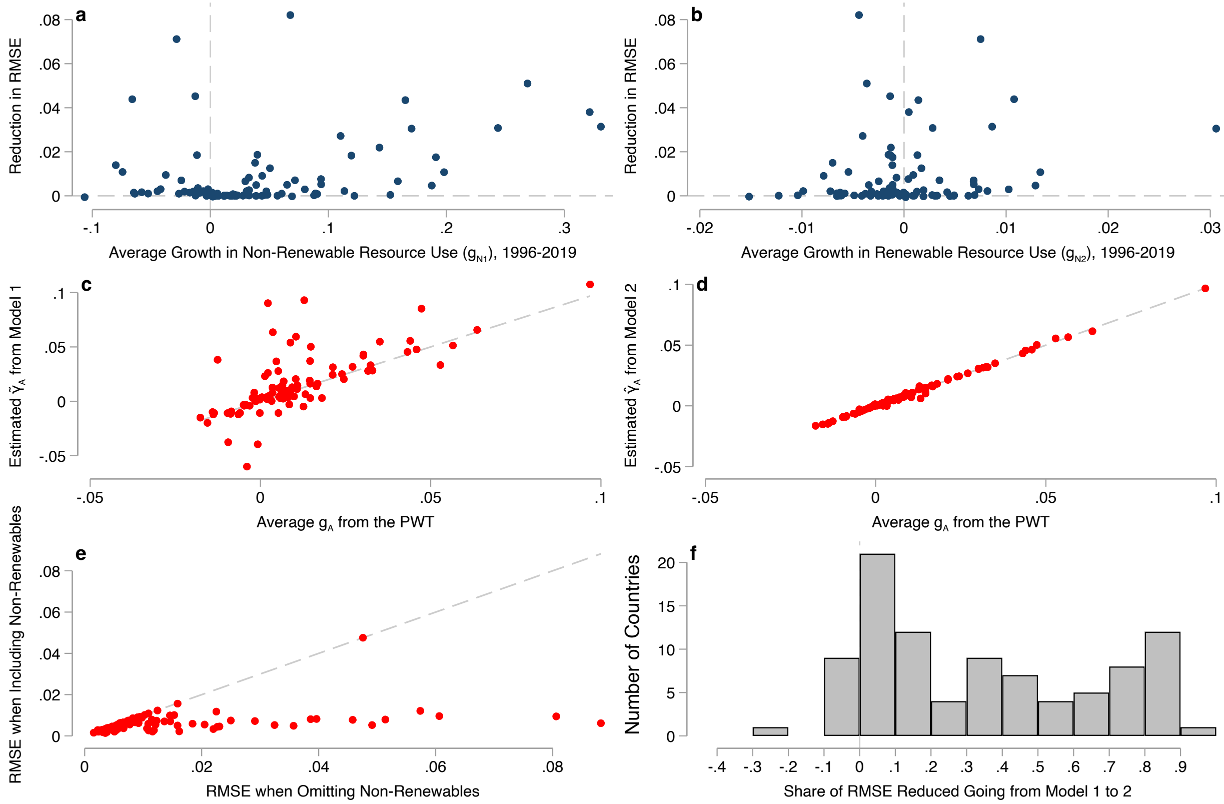

Figure 7: Panels 7a and 7b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 98 countries out of our original panel of 100 countries. Panels 7c and 7d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 7e and 7f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 7a, 7b, and 7f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 7c-7e show the 45 degree line.

fo

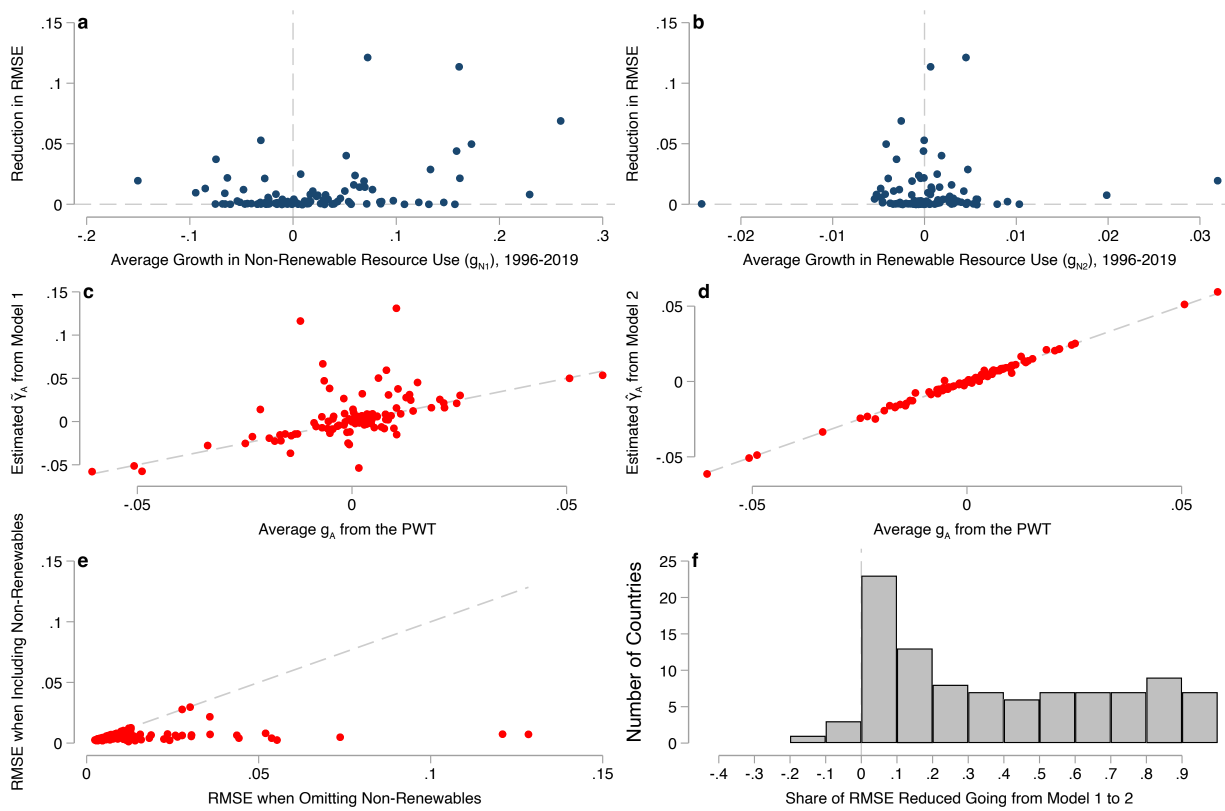

Figure 8: Panels 8a and 8b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for 98 countries out of our original panel of 100 countries. Panels 8c and 8d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 8e and 8f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 8a, 8b, and 8f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 8c-8e show the 45 degree line.

5 Hypothetical Fiscal Forecast Errors

The $300 billion figure referenced in Section 3 of the main text is constructed as follows. We assume that initial output in the United States is $30 trillion nominal USD in 2025 and grows at a rate of five percent each year, roughly in line with historical averages for the U.S. since 1990 (U.S. Bureau of Economic Analysis). We assume one percentage point of this growth is caused by growth in TFP, while inflation and growth in other factor inputs account for the remaining four points. This leads 10-year cumulative output between 2026 and 2035 of $396.2 trillion dollars. If an econometrician instead estimates that TFP will grow at a rate of 1.07 percent a year over the next ten years and correctly accounts for other drivers of nominal growth accounting for four points each year, she will forecast an average growth rate of 5.07 percent. This will generate a forecast of $397.7 trillion in cumulative output between 2026 and 2035. The difference in forecast output is $1.5 trillion. Assuming a ratio of tax receipts to GDP of roughly 20% (slightly above historical averages in the United States, see U.S. Office of Management and Budget and Federal Reserve Bank of St. Louis) this translates to an overshoot of (cumulative) revenue forecasts of $300 billion.

6 Proofs

6.1 Proposition 1

Part 1 Let $\bar{g} \triangleq \max\{\mathbb{E}[g_K], \mathbb{E}[g_L]\}$ and $\underline{g}_N \triangleq \min_j\{\mathbb{E}[g_{N_j}]\}$. Suppose that $\underline{g}_N > \bar{g}$. Then

$$\begin{aligned}
\mathbb{E}[\varepsilon] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1 - \hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&\geq \sum_{j=1}^{J} \gamma_j g_{N_j} - \sum_{j=1}^{J} \gamma_j \bar{g}
\end{aligned}$$

where the inequality follows from $\gamma_K + \gamma_L - 1 = \sum \gamma_j$. We then have

$$\mathbb{E}[\varepsilon] \geq [\underline{g}_N - \bar{g}] \sum_{j=1}^{J} \gamma_j \geq 0$$

Finally, note that adding a correctly-measured natural capital term lowers bias, as we reduce the number of positive factor shares γ_j in the summation in the last line which shrinks bias in absolute terms.

Part 2 Let $\underline{g} \triangleq \min\{g_K, g_L\}$ and $\bar{g}_N \triangleq \max_j\{g_{N_j}\}$ and suppose that $\underline{g} > \bar{g}_N$. Then analagous to the presentation above we have

$$\begin{aligned}
\mathbb{E}[\varepsilon] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1 - \hat{\gamma}_K)]\mathbb{E}[g_L] - \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&\leq \sum_{j=1}^{J} \gamma_j g_{N_j} - \sum_{j=1}^{J} \gamma_j \underline{g} \\
&\leq [\bar{g}_N - \underline{g}] \sum_{j=1}^{J} \gamma_j \leq 0
\end{aligned}$$

where in this case adding a correctly-measured NK stock (removing a term from the summation) decreases bias in nominal and absolute terms.

Part 3 Suppose $\hat{\gamma}_K = \gamma_K$. Then the bias is

$$\mathbb{E}[\varepsilon] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_L - g_{N_j}] \lesseqgtr 0$$

Alternatively, if $(1 - \hat{\gamma}_K) = \gamma_L$ then we have $\hat{\gamma}_K = \gamma_K + \sum \gamma$ which yields

$$\mathbb{E}[\varepsilon] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_K - g_{N_j}] \lesseqgtr 0$$

6.2 Lemma 1

Proof. We show here the result for a specific case where each input growth term is iid and output elasticities and factor input growth are measured and/or estimated without error. Introducing correlations in growth terms or measurement error adds more potential sources of ambiguity, but is not necessary to show that adding additional natural capital stocks has ambiguous effects on bias. In this case, the biases under models 1 and 2 from equations (Bias 1) and (Bias 2) become

$$\mathbb{E}[\varepsilon|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L)] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[\underbrace{g_{N_j}}_{\lesseqgtr 0}]$$

$$\mathbb{E}[\delta|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1)] = \sum_{j=2}^{J} \gamma_j \mathbb{E}[\underbrace{g_{N_j}}_{\lesseqgtr 0}]$$

and the magnitude and sign of both biases remains indeterminate.

The mean squared errors of our estimators when all factor shares and growth terms are measured without error are

$$\text{MSE}(1)|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L) = \mathbb{E}\left[\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right)^2\right]$$

$$\text{MSE}(2)|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1) = \mathbb{E}\left[\left(\sum_{j=2}^{J} \gamma_j g_{N_j}\right)^2\right]$$

Evaluating both means squared errors gives

$$\begin{aligned}
\text{MSE}(1)|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L) &= \mathbb{E}\left[\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right)^2\right] \\
&= \text{Var}\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right) + \mathbb{E}\left[\varepsilon\right]^2 \\
&= \sum_{i=1}^{J}\sum_{j=1}^{J} \gamma_j \gamma_i \text{Cov}\left(g_{N_i}, g_{N_j}\right) + \mathbb{E}\left[\varepsilon\right]^2
\end{aligned}$$

For the other estimator we have the analog less the included NK stock N_1:

$$\text{MSE}(2)|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1) = \sum_{i=2}^{J}\sum_{j=2}^{J} \gamma_j \gamma_i \text{Cov}\left(g_{N_i}, g_{N_j}\right) + \mathbb{E}\left[\delta\right]^2$$

The difference in mean squared error is

$$\text{MSE}(1) - \text{MSE}(2) = \gamma_1^2 \underbrace{\text{Var}(g_{N_1})}_{\geq 0} + \underbrace{2\gamma_1 \sum_{i=2}^{J} \gamma_j \text{Cov}\left(g_{N_1}, g_{N_j}\right)}_{\lesseqgtr 0} + \underbrace{\mathbb{E}[\varepsilon]^2 - \mathbb{E}[\delta]^2}_{\lesseqgtr 0} \lesseqgtr 0$$

Under the assumption of iid growth terms for all natural capital inputs g_{N_j}, the above difference becomes

$$\text{MSE}(1) - \text{MSE}(2) = \gamma_1^2 \underbrace{\text{Var}(g_{N_1})}_{\geq 0} + \underbrace{\mathbb{E}[\varepsilon]^2 - \mathbb{E}[\delta]^2}_{\lesseqgtr 0} \lesseqgtr 0$$

where the inequality is ambiguous unless $\mathbb{E}[\delta] = 0$ in the case where including N_1 makes the growth accounting equation correctly specified.

□

6.3 Proposition 2

Proof. The result follows directly from equations (12) and (13) as we've assumed that factor growth terms are independent random variables. □

6.4 Biases of OLS-Based TFP Estimators

We present the bias calculations here for the general case where the set of natural capital stocks has cardinality $J \geq 2$. The bias in expected TFP growth when factor shares are estimated using specification in (20) with OLS

$$
\begin{aligned}
\mathbb{E}[\hat{g}_A - \mathbb{E}[g_{TFP}]] &= \mathbb{E}[g_y] - \mathbb{E}[g_y] + [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&= -\left[\sum_{j=1}^{J} \gamma_j \kappa_{N_j,\tilde{K}} + \kappa_{A,\tilde{K}}\right]\mathbb{E}[g_K] - \left[\sum_{j=1}^{J} \gamma_j \kappa_{N_j,\tilde{L}} + \kappa_{A,\tilde{L}}\right]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&= \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j} - \kappa_{N_j,\tilde{K}} g_K - \kappa_{N_j,\tilde{L}} g_L] - \kappa_{A,\tilde{K}}\mathbb{E}[g_K] - \kappa_{A,\tilde{L}}\mathbb{E}[g_L]
\end{aligned}
$$

where the κ_{N_j} coefficients are generated by linear projections of g_{N_j} onto the vector $[g_K, g_L]'$. The bias generated by the OLS specification in (21) follows in a similar manner

$$
\begin{aligned}
\mathbb{E}[\tilde{g}_A - \mathbb{E}[g_{TFP}]] = &-\left[\sum_{j=2}^{J} \gamma_j \lambda_{N_j,\tilde{K}} + \lambda_{A,\tilde{K}}\right]\mathbb{E}[g_K] - \left[\sum_{j=2}^{J} \gamma_j \lambda_{N_j,\tilde{L}} + \lambda_{A,\tilde{L}}\right]\mathbb{E}[g_L] \\
&-\left[\sum_{j=2}^{J} \gamma_j \lambda_{N_j,\tilde{N}_1} + \lambda_{A,\tilde{N}_1}\right]\mathbb{E}[g_{N_1}] + \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&= \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j} - \lambda_{N_j,\tilde{K}} g_K - \lambda_{N_j,\tilde{L}} g_L - \lambda_{N_j,\tilde{N}_1} g_{N_1}] \\
&- \lambda_{A,\tilde{K}}\mathbb{E}[g_K] - \lambda_{A,\tilde{L}}\mathbb{E}[g_L] - \lambda_{A,\tilde{N}_1}\mathbb{E}[g_{N_1}]
\end{aligned}
$$

where λ_{N_2} are coefficients from a projection of g_{N_2} onto $[g_K, g_L, g_{N_1}]$.

6.5 Variance the of OLS-Based TFP Estimators

Model 1 Split the true data generating process from (19) into vectors of included and excluded regressors

$$
y = Z\gamma_Z + X_o\gamma_o + \varepsilon \tag{26}
$$

where

$$
Z = [\mathbf{1}, g_K, g_L], \qquad X_o = [g_{N_1}, g_{N_2}],
$$

and

$$
\gamma_Z = (\mathbb{E}[g_{TFP}], \gamma_K, \gamma_L)^\top, \qquad \gamma_o = (\gamma_1, \gamma_2)^\top
$$

We assume here that the error term ε—the portion of output growth unexplained by input growth each period—is governed by residual variation in TFP growth such that

$$\mathbb{E}[\varepsilon|\boldsymbol{g}] = 0 \qquad \mathrm{Var}(\varepsilon) = \sigma^2_{TFP}$$

The OLS estimator from regressing y on Z alone is

$$\hat{\gamma} = (Z'Z)^{-1}Z'y. \tag{27}$$

Substituting the true model,

$$\hat{\gamma} = (Z'Z)^{-1}Z'(Z\gamma_Z + X_o\gamma_o + \varepsilon) \tag{28}$$

$$= \gamma_Z + (Z'Z)^{-1}Z'X_o\gamma_o + (Z'Z)^{-1}Z'\varepsilon\gamma_o \tag{29}$$

Define the combined error term as

$$u^{(1)} \triangleq X_o\gamma_o + \varepsilon \tag{30}$$

The variance of $\hat{\gamma}$ conditional on Z is given by the traditional sandwich estimand

$$\mathrm{Var}(\hat{\gamma} \mid Z) = (Z'Z)^{-1}Z'\Omega_1 Z(Z'Z)^{-1} \tag{31}$$

where

$$\Omega_1 \triangleq \mathrm{Var}(u^{(1)} \mid Z)$$

is a covariance matrix of size $n = 25$. Under the assumption of homoskedastic errors and independent sampling we can write the covariance matrix Ω_1 as

$$\Omega_1 = \tau_1^2 I_n$$

where I_n is the identity matrix of size equal to the sample size n and τ_1^2 is a function of the variance of TFP growth σ^2_{TFP} along with the variance of the omitted terms

$$\tau_1^2 = \sigma^2_{TFP} + \mathrm{Var}(\gamma_1 g_{N_1} + \gamma_2 g_{N_2} \mid Z) \tag{32}$$

Letting

$$\Sigma_{12|Z} \triangleq \mathrm{Var}\left(\begin{bmatrix} g_{N_1} \\ g_{N_2} \end{bmatrix} \,\middle|\, Z\right)$$

we can write the variance term as

$$\mathrm{Var}(\gamma_1 g_{N_1} + \gamma_2 g_{N_2} \mid Z) = \gamma_o^\top \Sigma_{12|Z}\gamma_o \tag{33}$$

Which gives the result in Section 4.3:

$$\text{Var}(\hat{\gamma} \mid Z) = \left(\sigma^2_{TFP} + \gamma_o^\top \Sigma_{12\cdot Z}\gamma_o\right)(Z'Z)^{-1} \tag{34}$$

Model 2 Rewrite the true data generating process from (DGP) as

$$y = W\gamma_W + g_{N_2}\gamma_2 + \varepsilon \tag{35}$$

where, similar to the procedure for model 1, we assume

$$\mathbb{E}[\varepsilon|\boldsymbol{g}] = 0 \qquad \text{Var}(\varepsilon) = \sigma^2_{TFP} \qquad W = [\mathbf{1}, g_K, g_L, g_{N_1}], \qquad \gamma_W = (\mathbb{E}[g_{TFP}], \gamma_K, \gamma_L, \gamma_1)^\top.$$

The OLS estimand from regressing y on W is

$$\tilde{\gamma} = (W'W)^{-1}W'y. \tag{36}$$

Substituting gives

$$\tilde{\gamma} = (W'W)^{-1}W'(W\gamma_W + g_{N_2}\gamma_2) \tag{37}$$

$$= \gamma_W + (W'W)^{-1}W'g_{N_2}\gamma_2 + (W'W)^{-1}W'\varepsilon \tag{38}$$

Define the error for model 2 as

$$u^{(2)} \triangleq \gamma_2 g_{N_2} + \varepsilon \tag{39}$$

Then the expected variance conditional on W is

$$\text{Var}(\tilde{\gamma} \mid W) = (W'W)^{-1}W'\Omega_2 W(W'W)^{-1} \tag{40}$$

where

$$\Omega_2 \triangleq \text{Var}(u^{(2)} \mid W).$$

Under the assumption of homoskedastic errors and independent sampling we can write the covariance matrix Ω_2 as

$$\Omega_2 = \tau_2^2 I_n,$$

where (analagous to above)

$$\tau_2^2 = \sigma^2_{TFP} + \gamma_2^2 \text{Var}(g_{N_2} \mid W). \tag{41}$$

Substituting gives

$$\mathrm{Var}(\tilde{\gamma} \mid W) = \left(\sigma^2 + \gamma_2^2 \mathrm{Var}(g_{N_2} \mid W)\right)(W'W)^{-1} \tag{42}$$

as written in Section 4.3.

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